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Board connector

Abstract

The present disclosure relates to a board connector comprising a plurality of RF contacts; an insulation unit; a plurality of transmit contacts and a plurality of second RF contacts, so that the first and second RF contacts are spaced apart along a first axial direction; a ground housing; a first ground contact coupled to the insulation unit, and providing shielding between the first RF contacts and the transmit contacts, with respect to the first axial direction; and a second ground contact coupled to the insulation unit, and providing shielding between the second RF contacts and the transmit contacts, with respect to the first axial direction, wherein the first ground contact provides shielding between the first RF contacts and the transmit contacts, with respect to the first axial direction, and provides shielding between the first RF contacts with respect to a second axial direction perpendicular to the first axial direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is a National Stage of International Application No. PCT/KR2021/002843 filed on Mar. 8, 2021, which claims priority to and the benefit of Korean Utility Model Application No. 10-2020-0033572, filed on Mar. 19, 2020; and Korean Utility Model Application No. 10-2021-0029518, filed Mar. 5, 2021 the disclosures of which are incorporated herein by reference in their entirety.

FIELD

(2) The present disclosure relates to a board connector installed in an electronic device for electrical connection between boards.

BACKGROUND

(3) Connectors are provided in various electronic devices for electrical connection. For example, the connectors may be installed in electronic devices such as mobile phones, computers, tablet computers, and the like to electrically connect various components installed in the electronic devices to each other.

(4) In general, radio frequency (RF) connectors and board-to-board connectors (hereinafter, referred to as "board connectors") are provided inside wireless communication devices such as smartphones, tablet personnel computers (PCs), or the like among the electronic devices. The RF connectors transmit RF signals. The board connectors process digital signals of cameras or the like.

(5) Such an RF connector and board connector are mounted on a printed circuit board (PCB). Conventionally, there is a problem that a mounting area of a PCB increases since a plurality of board connectors and RF connectors are mounted in a limited PCB space together with a plurality of components. Accordingly, with a recent trend of miniaturization of smartphones, there is a need for a technique in which the RF connector and the board connector are integrated and optimized for a small mounting area on the PCB.

(6) FIG. **1** is a schematic perspective view of a board connector according to the related art.

(7) Referring to FIG. **1**, a board connector **100** according to the related art includes a first connector **110** and a second connector **120**.

(8) The first connector **110** is provided to be coupled to a first board (not shown). The first connector **110** may be electrically connected to the second connector **120** through a plurality of first contacts **111**.

(9) The second connector **120** is provided to be coupled to a second board (not shown). The second connector **120** may be electrically connected to the first connector **110** through a plurality of second contacts **121**.

(10) In the board connector **100** according to the related art, as the first contacts **111** and the second contacts **121** are connected to each other, the first board and the second board may be electrically connected to each other. In addition, when some contacts among the first contacts **111** and the second contacts **121** are used as RF contacts for transmitting RF signals, the board connector **100** according to the related art may be realized such that the RF signals are transmitted between the first board and the second board through the RF contacts.

(11) Here, the board connector **100** according to the related art has the following problems.

(12) First, in the board connector **100** according to the related art, when contacts, which are spaced apart by a relatively small distance, among the contacts **111** and **121** are used as the RF contacts, there is a problem in that signal transmission is not smoothly performed due to RF signal interference between RF contacts **111′** and **111″** and between RF contacts **121′** and **121″**.

(13) Second, the board connector **100** according to the related art has an RF signal shielding unit **112** at an outermost portion thereof, and thus there is a problem in that radiation of the RF signal to the outside can be shielded but shielding between the RF signals is not performed.

(14) Third, in the board connector **100** according to the related art, the RF contacts **111′**, **111″**, **121′**, and **121″** respectively include mounting units **111**_a_′, **111**_a_″, **121**_a_′, and **121**_a_″ mounted on the board,

and the mounting units **111***a*′, **111***a*″, **121***a*′, and **121***a*″ are disposed to be exposed to the outside. Accordingly, there is a problem in that shielding for the mounting units **111***a*′, **111***a*″, **121***a*′, and **121***a*″ is not performed in the board connector **100** according to the related art.

SUMMARY

(15) Therefore, the present disclosure is designed to solve the problems and is for providing a board connector capable of reducing the possibility of occurring radio frequency (RF) signal interference between RF contacts.

(16) To solve the above problems, the present disclosure may include the following configurations.

(17) A board connector according to the present disclosure may include a plurality of radio frequency (RF) contacts for transmitting an RF signal, an insulation unit configured to support the RF contacts, a plurality of transmit contacts that are coupled to the insulation unit between a plurality of first RF contacts among the RF contacts and a plurality of second RF contacts among the RF contacts such that the first RF contacts and the second RF contacts are spaced apart from each other along a first axial direction, a ground housing to which the insulation unit is coupled, a first ground contact coupled to the insulation unit and configured to shield between the first RF contacts and the transmit contacts with respect to the first axial direction, and a second ground contact coupled to the insulation unit and configured to shield between the second RF contacts and the transmit contacts with respect to the first axial direction. The first ground contact may shield between the first RF contacts and the transmit contacts with respect to the first axial direction, and shield between the first RF contacts with respect to a second axial direction perpendicular to the first axial direction.

(18) A board connector according to the present disclosure may include a plurality of radio frequency (RF) contacts for transmitting an RF signal, an insulation unit configured to support the RF contacts, a plurality of transmit contacts that are coupled to the insulation unit between a plurality of first RF contacts among the RF contacts and a plurality of second RF contacts among the RF contacts such that the first RF contacts and the second RF contacts are spaced apart from each other along a first axial direction, a ground housing to which the insulation unit is coupled, a first ground contact coupled to the insulation unit and configured to shield between the first RF contacts and the transmit contacts with respect to the first axial direction, and a second ground contact coupled to the insulation unit and configured to shield between the second RF contacts and the transmit contacts with respect to the first axial direction.

(19) According to the present disclosure, the following effects can be obtained.

(20) The present disclosure can realize a shielding function against signals, electromagnetic waves, or the like for radio frequency (RF) contacts using a ground housing and a ground contact. Thus, the present disclosure can prevent electromagnetic waves, which are generated from the RF contacts, from interfering with signals of circuit components located around an electronic device, and prevent electromagnetic waves, which are generated from the circuit components located around the electronic device, from interfering with RF signals transmitted by the RF contacts. Accordingly, the present disclosure can contribute to improving electromagnetic interference (EMI) shielding performance and electromagnetic compatibility (EMC) performance using the ground housing and the ground contact.

(21) In the present disclosure, it can be realized such that all RF contacts, including portions mounted on a board, are located inside a ground housing. Accordingly, the present disclosure can realize complete shielding by enhancing a shielding function for the RF contacts using the ground housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic perspective view of a board connector according to the related art.

(2) FIG. **2** is a schematic perspective view of a receptacle connector and a plug connector in a board connector according to the present disclosure.

(3) FIG. **3** is a schematic perspective view of a board connector according to a first embodiment.

(4) FIG. **4** is a schematic exploded perspective view of the board connector according to the first embodiment.

(5) FIG. **5** is a schematic plan view for describing a ground loop in the board connector according to the first embodiment.

(6) FIG. **6** is a schematic perspective view of a first ground contact and a second ground contact in the board connector according to the first embodiment.

(7) FIG. **7** is a schematic plan view of the board connector according to the first embodiment.

(8) FIG. **8** is a schematic cross-sectional side view taken along line I-I of FIG. **7**, illustrating a state in which the board connector according to the first embodiment is coupled to a board connector according to a second embodiment.

(9) FIG. **9** is a schematic cross-sectional side view taken along line II-II of FIG. **7**, illustrating the state in which the board connector according to the first embodiment is coupled to the board connector according to the second embodiment.

(10) FIG. **10** is a schematic perspective view of a ground housing in the board connector according to the first embodiment.

(11) FIGS. **11** to **14** are enlarged schematic cross-sectional side views of portion A of FIG. **8**, illustrating the state in which the board connector according to the first embodiment is coupled to the board connector according to the second embodiment.

(12) FIG. **15** is a schematic perspective view of a modified embodiment of the ground housing in the board connector according to the first embodiment.

(13) FIG. **16** is a schematic plan view of an insulation unit in the board connector according to the first embodiment.

(14) FIG. **17** is a schematic perspective view of the board connector according to the second embodiment.

(15) FIG. **18** is a schematic exploded perspective view of the board connector according to the second embodiment.

(16) FIG. **19** is a schematic plan view of the board connector according to the second embodiment.

(17) FIG. **20** is a schematic cross-sectional side view taken along line of FIG. **19**, illustrating the state in which the board connector according to the second embodiment is coupled to the board connector according to the first embodiment.

(18) FIG. **21** is a schematic plan view for describing a ground loop in the board connector according to the second embodiment.

(19) FIG. **22** is a schematic perspective view of a ground housing in the board connector according to the second embodiment.

(20) FIG. **23** is an enlarged schematic cross-sectional side view of portion A of FIG. **8**, illustrating the state in which the board connector according to the second embodiment is coupled to the board connector according to the first embodiment.

(21) FIGS. **24** to **27** are conceptual bottom views illustrating an embodiment of a mounting pattern of a board, on which the board connector according to the first embodiment is mounted.

(22) FIGS. **28** to **31** are conceptual bottom views illustrating an embodiment of a mounting pattern of a board, on which the board connector according to the second embodiment is mounted.

DETAILED DESCRIPTION

(23) Hereinafter, embodiments of a board connector according to the present disclosure will be described in detail with reference to the accompanying drawings. FIGS. **8** and **9** illustrate a state in

which a connector according to a first embodiment is reversed in a direction shown in FIGS. **2** and **3** and coupled to a connector according to a second embodiment.

(24) Referring to FIG. **2**, a board connector **1** according to the present disclosure may be installed in an electronic device (not shown) such as a mobile phone, a computer, a tablet computer, or the like. The board connector **1** according to the present disclosure may be used to electrically connect a plurality of boards (not shown). The boards may be printed circuit boards (PCBs). For example, when a first board and a second board are electrically connected, a receptacle connector mounted on the first board and a plug connector mounted on the second board may be connected to each other. Accordingly, the first board and the second board may be electrically connected to each other through the receptacle connector and the plug connector. A plug connector mounted on the first board and a receptacle connector mounted on the second board may also be connected to each other.

(25) The board connector **1** according to the present disclosure may be realized as the receptacle connector. The board connector **1** according to the present disclosure may be realized as the plug connector. The board connector **1** according to the present disclosure may also be realized by including both the receptacle connector and the plug connector. Hereinafter, the board connector according to an embodiment in which the board connector **1** according to the present disclosure is realized as the plug connector is defined as a board connector **200** according to the first embodiment, and the board connector according to an embodiment in which the board connector **1** according to the present disclosure is realized as the receptacle connector is defined as a board connector **300** according to the second embodiment, and these will be described in detail with reference to the accompanying drawings. In addition, descriptions will be made on the basis of an embodiment in which the board connector **200** according to the first embodiment is mounted on the first board and the board connector **300** according to the second embodiment is mounted on the second board. It should be apparent to those skilled in the art to which the present disclosure belongs to derive an embodiment, in which the board connector **1** according to the present disclosure includes both the receptacle connector and the plug connector, therefrom.

(26) <Board Connector **200** According to First Embodiment>

(27) Referring to FIGS. **2** to **4**, the board connector **200** according to the first embodiment may include a plurality of radio frequency (RF) contacts **210**, a plurality of transmit contacts **220**, a ground housing **230**, and an insulation unit **240**.

(28) The RF contacts **210** are for transmitting RF signals. The RF contacts **210** may transmit ultra-high frequency RF signals. The RF contacts **210** may be supported by the insulation unit **240**. The RF contacts **210** may be coupled to the insulation unit **240** through an assembly process. The RF contacts **210** may also be integrally molded with the insulation unit **240** through injection molding.

(29) The RF contacts **210** may be disposed to be spaced apart from each other. The RF contacts **210** may be electrically connected to the first board by being mounted on the first board. The RF contacts **210** may be electrically connected to the second board, on which a mating connector is mounted, by being connected to RF contacts included in the mating connector. Accordingly, the first board and the second board may be electrically connected. When the board connector **200** according to the first embodiment is a plug connector, the mating connector may be a receptacle connector. When the board connector **200** according to the first embodiment is a receptacle connector, the mating connector may be a plug connector.

(30) A first RF contact **211** among the RF contacts **210** and a second RF contact **212** among the RF contacts **210** may be spaced apart from each other along a first axial direction (X-axis direction). The first RF contact **211** and the second RF contact **212** may be supported by the insulation unit **240** at locations spaced apart from each other along the first axial direction (X-axis direction).

(31) The first RF contact **211** may include a first RF mounting member **2111**. The first RF mounting member **2111** may be mounted on the first board. Accordingly, the first RF contact **211** may be electrically connected to the first board through the first RF mounting member **2111**. The

first RF contact **211** may be formed of an electrically conductive material. For example, the first RF contact **211** may be formed of metal. The first RF contact **211** may be connected to any one of the RF contacts included in the mating connector.

(32) The second RF contact **212** may include a second RF mounting member **2121**. The second RF mounting member **2121** may be mounted on the first board. Accordingly, the second RF contact **212** may be electrically connected to the first board through the second RF mounting member **2121**. The second RF contact **212** may be formed of an electrically conductive material. For example, the second RF contact **212** may be formed of metal. The second RF contact **212** may be connected to any one of the RF contacts included in the mating connector.

(33) Referring to FIGS. **2** to **4**, the transmit contacts **220** are coupled to the insulation unit **240**. The transmit contacts **220** may serve to transmit signals, data, and the like. The transmit contacts **220** may be coupled to the insulation unit **240** through an assembly process. The transmit contacts **220** may also be integrally molded with the insulation unit **240** through injection molding.

(34) The transmit contacts **220** may be disposed between the first RF contact **211** and the second RF contact **212** with respect to the first axial direction (X-axis direction). Accordingly, in order to reduce RF signal interference between the first RF contact **211** and the second RF contact **212**, the transmit contacts **220** may be disposed in a space in which the first RF contact **211** and the second RF contact **212** are spaced apart. Accordingly, the board connector **200** according to the first embodiment may not only reduce RF signal interference by increasing a distance by which the first RF contact **211** and the second RF contact **212** are spaced apart from each other, but also improve space utilization for the insulation unit **240** by disposing the transmit contacts **220** in a separation space for this purpose.

(35) The transmit contacts **220** may be disposed to be spaced apart from each other. The transmit contacts **220** may be electrically connected to the first board by being mounted on the first board. In this case, a transmission mounting member **2201** included in each of the transmit contacts **220** may be mounted on the first board. The transmit contacts **220** may be formed of an electrically conductive material. For example, the transmit contacts **220** may be formed of metal. The transmit contacts **220** may be electrically connected to the second board, on which the mating connector is mounted, by being connected to transmit contacts included in the mating connector. Accordingly, the first board and the second board may be electrically connected.

(36) Meanwhile, in FIG. **4**, the board connector **200** according to the first embodiment is illustrated as including four transmit contacts **220**, but the present disclosure is not limited thereto, and the board connector **200** according to the first embodiment may also include five or more transmit contacts **220**. The transmit contacts **220** may be spaced apart from each other along the first axial direction (X-axis direction) and a second axial direction (Y-axis direction). The first axial direction (X-axis direction) and the second axial direction (Y-axis direction) are axis directions perpendicular to each other.

(37) Referring to FIGS. **2** to **4**, the ground housing **230** is coupled to the insulation unit **240**. The ground housing **230** may be grounded by being mounted on the first board. Accordingly, the ground housing **230** may realize a shielding function against signals, electromagnetic waves, or the like for the RF contacts **210**. In this case, the ground housing **230** may prevent electromagnetic waves generated from the RF contacts **210** from interfering with signals of circuit components located around the electronic device, and may prevent electromagnetic waves generated from the circuit components located around the electronic device from interfering with RF signals transmitted by the RF contacts **210**. Accordingly, the board connector **200** according to the first embodiment may contribute to improving electromagnetic interference (EMI) shielding performance and electromagnetic compatibility (EMC) performance using the ground housing **230**. The ground housing **230** may be formed of an electrically conductive material. For example, the ground housing **230** may be formed of metal.

(38) The ground housing **230** may be disposed to surround sides of an inner side space **230*a***. A

portion of the insulation unit **240** may be located in the inner side space **230***a*. All of the first RF contact **211**, the second RF contact **212**, and the transmit contacts **220** may be located in the inner side space **230***a*. In this case, all of the first RF mounting member **2111**, the second RF mounting member **2121**, and the transmission mounting member **2201** may also be located in the inner side space **230***a*. Accordingly, the ground housing **230** may enhance a shielding function for the first RF contact **211** and the second RF contact **212** by realizing shielding walls for all of the first RF contact **211** and the second RF contact **212**, thereby realizing complete shielding. The mating connector may be inserted into the inner side space **230***a*.
(39) The ground housing **230** may be disposed to surround all sides of the inner side space **230***a*. The inner side space **230***a* may be disposed inside the ground housing **230**. When the entire ground housing **230** is formed in a rectangular loop shape, the inner side space **230***a* may be formed in a rectangular parallelepiped shape. In this case, the ground housing **230** may be disposed to surround four sides of the inner side space **230***a*.
(40) The ground housing **230** may be integrally formed as one piece without a seam. The ground housing **230** may be integrally formed as one piece without a seam by a metal injection method, such as a metal die casting method, a metal injection molding (MIM) method, or the like. The ground housing **230** may be integrally formed as one piece without a seam by a computer numerical control (CNC) process, a machining center tool (MCT) process, or the like.
(41) Referring to FIGS. **2** to **4**, the insulation unit **240** supports the RF contacts **210**. The RF contacts **210** and the transmit contacts **220** may be coupled to the insulation unit **240**. The insulation unit **240** may be formed of an insulating material. The insulation unit **240** may be coupled to the ground housing **230** such that the RF contacts **210** are located in the inner side space **230***a*.
(42) Referring to FIGS. **2** to **4**, the board connector **200** according to the first embodiment may include a first ground contact **250**.
(43) The first ground contact **250** is coupled to the insulation unit **240**. The first ground contact **250** may be grounded by being mounted on the first board. The first ground contact **250** may be coupled to the insulation unit **240** through an assembly process. The first ground contact **250** may also be integrally molded with the insulation unit **240** through injection molding.
(44) The first ground contact **250** may realize a shielding function for the first RF contact **211** together with the ground housing **230**. In this case, the first ground contact **250** may be disposed between the first RF contact **211** and the transmit contacts **220** with respect to the first axial direction (X-axis direction). The first ground contact **250** may be formed of an electrically conductive material. For example, the first ground contact **250** may be formed of metal. When the mating connector is inserted into the inner side space **230***a*, the first ground contact **250** may be connected to a ground contact included in the mating connector.
(45) Referring to FIGS. **2** to **4**, the board connector **200** according to the first embodiment may include a second ground contact **260**.
(46) The second ground contact **260** is coupled to the insulation unit **240**. The second ground contact **260** may be grounded by being mounted on the first board. The second ground contact **260** may be coupled to the insulation unit **240** through an assembly process. The second ground contact **260** may also be integrally molded with the insulation unit **240** through injection molding.
(47) The second ground contact **260** may realize a shielding function for the second RF contact **212** together with the ground housing **230**. The second ground contact **260** may be disposed between the transmit contacts **220** and the second RF contact **212** with respect to the first axial direction (X-axis direction). The second ground contact **260** may be formed of an electrically conductive material. For example, the second ground contact **260** may be formed of metal. When the mating connector is inserted into the inner side space **230***a*, the second ground contact **260** may be connected to the ground contact included in the mating connector.
(48) Here, the board connector **200** according to the first embodiment may be realized to include a

plurality of first RF contacts **211** and a plurality of second RF contacts **212**.
(49) Referring to FIGS. **2** to **9**, the first RF contacts **211** and the second RF contacts **212** may be disposed to be spaced apart from each other along the first axial direction (X-axis direction). The transmit contacts **220** may be disposed between the first RF contacts **211** and the second RF contacts **212** with respect to the first axial direction (X-axis direction). In this case, the first ground contact **250** may shield between the first RF contacts **211** and the transmit contacts **220** with respect to the first axial direction (X-axis direction). The second ground contact **260** may shield between the second RF contacts **212** and the transmit contacts **220** with respect to the first axial direction (X-axis direction).
(50) When the plurality of first RF contacts **211** are provided, the first ground contact **250** may shield between the first RF contacts **211** and the transmit contacts **220** with respect to the first axial direction (X-axis direction), and also, shield between the first RF contacts **211** with respect to the second axial direction (Y-axis direction). Accordingly, by using the first ground contact **250**, the board connector **200** according to the first embodiment may realize a shielding function for between the first RF contacts **211** and the transmit contacts **220** and also, additionally realize a shielding function for between the first RF contacts **211**. Accordingly, the board connector **200** according to the first embodiment may be realized to transmit a wider variety of RF signals using the first RF contacts **211**, thereby improving versatility applicable to a wider variety of electronic products.
(51) A first-first RF contact **211***a* among the first RF contacts **211** and a first-second RF contact **211***b* among the first RF contacts **211** may be coupled to the insulation unit **240** so as to be spaced apart from each other along the second axial direction (Y-axis direction). In FIG. **5**, the board connector **200** according to the first embodiment is illustrated as including two first RF contacts **211** realized as the first-first RF contact **211***a* and the first-second RF contact **211***b*, but the present disclosure is not limited thereto, and the board connector **200** according to the first embodiment may also include three or more first RF contacts **211**. Meanwhile, in the present specification, descriptions will be made on the basis of the case in which the board connector **200** according to the first embodiment includes the first-first RF contact **211***a* and the first-second RF contact **211***b*.
(52) When the first-first RF contact **211***a* and the first-second RF contact **211***b* are provided, the first ground contact **250** may include a first-first ground contact **251** and a first-second ground contact **252**.
(53) The first-first ground contact **251** may be located between the first-first RF contact **211***a* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). Accordingly, the first-first ground contact **251** may shield between the first-first RF contact **211***a* and the transmit contacts **220**.
(54) The first-first ground contact **251** may include a first-first shield member **2511**.
(55) The first-first shield member **2511** may be located between the first-first RF contact **211***a* and the first-second RF contact **211***b* with respect to the second axial direction (Y-axis direction). Accordingly, the first-first ground contact **251** may shield between the first-first RF contact **211***a* and the first-second RF contact **211***b* using the first-first shield member **2511**. Accordingly, even though the first-first RF contact **211***a* and the first-second RF contact **211***b* transmit different RF signals, the board connector **200** according to the first embodiment may prevent signals or the like from being interfered between the first-first RF contact **211***a* and the first-second RF contact **211***b* using the first-first shield member **2511**. Accordingly, the board connector **200** according to the first embodiment may be realized to stably transmit a wider variety of RF signals using the first-first RF contact **211***a* and the first-second RF contact **211***b*. The first-first shield member **2511** may be formed in a plate shape disposed in a vertical direction between the first-first RF contact **211***a* and the first-second RF contact **211***b*.
(56) The first-first shield member **2511** may be disposed to be spaced apart from each of the first-first RF contact **211***a* and the first-second RF contact **211***b* with respect to the second axial

direction (Y-axis direction) by the same distance. Accordingly, in the board connector **200** according to the first embodiment, a deviation between shielding performance for the first-first RF contact **211***a* and shielding performance for the first-second RF contact **211***b* may be reduced. Accordingly, the board connector **200** according to the first embodiment may stably realize a shielding function for each of the first-first RF contact **211***a* and the first-second RF contact **211***b* using the first-first shield member **2511**.
(57) The first-first ground contact **251** may include a first-first shield protrusion **2512**.
(58) The first-first shield protrusion **2512** protrudes from the first-first shield member **2511**. The first-first shield protrusion **2512** may be connected to the ground housing **230**. Accordingly, the first ground contact **250** may enhance shielding performance between the first-first RF contact **211***a* and the first-second RF contact **211***b* by being electrically connected to the ground housing **230** through the first-first shield protrusion **2512**, thereby realizing complete shielding. The first-first shield protrusion **2512** may be formed in a plate shape disposed in the vertical direction.
(59) The first-first ground contact **251** may include a first-first ground connection member **2513** and a first-first ground mounting member **2514**.
(60) The first-first ground connection member **2513** is coupled to each of the first-first shield member **2511** and the first-first ground mounting member **2514**. The first-first shield member **2511** and the first-first ground mounting member **2514** may be connected to each other through the first-first ground connection member **2513**. The first-first ground connection member **2513** may be connected to the ground contact included in the mating connector. Accordingly, the first ground contact **250** may be electrically connected to the ground contact included in the mating connector by being connected to the ground contact included in the mating connector through the first-first ground connection member **2513**. Accordingly, a gap generated as the first-first ground contact **251** and the first-second ground contact **252** are disposed to be spaced apart from each other along the second axial direction (Y-axis direction) may be shielded as the first ground contact **250** is connected to the ground contact included in the mating connector through the first-first ground connection member **2513**. The first-first shield member **2511** may be coupled to the first-first ground connection member **2513**. The first-first shield member **2511** may protrude from the first-first ground connection member **2513** along the first axial direction (X-axis direction). In this case, the first-first shield protrusion **2512** may protrude from the first-first shield member **2511** along the first axial direction (X-axis direction).
(61) The first-first ground mounting member **2514** is mounted on the first board. The first-first ground mounting member **2514** may be grounded by being mounted on the first board. Accordingly, the first-first ground contact **251** may be grounded to the first board through the first-first ground mounting member **2514**. The first-first ground mounting member **2514** may protrude from the first-first ground connection member **2513** along the second axial direction (Y-axis direction). In this case, the first-first ground mounting member **2514** may be disposed between the first-first RF contact **211***a* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). The first-first ground mounting member **2514** may protrude from the first-first ground connection member **2513** by a length connectable to the ground housing **230** with respect to the second axial direction (Y-axis direction). In this case, the first-first ground mounting member **2514** and the first-first shield member **2511** protrude from the first-first ground connection member **2513** in different directions to be connected to different sidewalls included in the ground housing **230**. Accordingly, since the first-first ground contact **251** and the ground housing **230** are electrically connected to each other while surrounding all sides of the first-first RF contact **211***a*, the board connector **200** according to the first embodiment may further enhance the shielding performance for the first-first RF contact **211***a*, thereby realizing complete shielding. The first-first ground mounting member **2514** may be formed in a plate shape disposed in a horizontal direction.
(62) The first-first ground contact **251** may include a first-first ground protrusion **2515**.
(63) The first-first ground protrusion **2515** protrudes from the first-first shield member **2511**. The

first-first ground protrusion **2515** may be mounted on the first board. Accordingly, a mounting area in which the first-first ground contact **251** is mounted on the first board may be increased, so that the board connector **200** according to the first embodiment may further enhance the shielding performance using the first-first ground contact **251**. The first-first ground protrusion **2515** may be mounted on the first board by passing through the insulation unit **240** and protruding from the insulation unit **240**. The first-first ground protrusion **2515** may protrude from the first-first shield member **2511** along the vertical direction. The first-first ground protrusion **2515** may be formed in a plate shape disposed in the vertical direction.

(64) The first-first ground contact **251** may include a first-first connection protrusion **2516**.

(65) The first-first connection protrusion **2516** protrudes from the first-first shield member **2511**. The first-first connection protrusion **2516** may be connected to a ground housing of the mating connector. Accordingly, a connection area in which the first-first ground contact **251** is connected to the ground housing of the mating connector may be increased, so that the board connector **200** according to the first embodiment may further enhance the shielding performance using the first-first ground contact **251**. The first-first connection protrusion **2516** may be connected to the ground housing of the mating connector by passing through the insulation unit **240** and protruding from the insulation unit **240**. The first-first connection protrusion **2516** may be inserted into an insulation unit included in the mating connector to be connected to the ground housing included in the mating connector. In this case, a through hole into which the first-first connection protrusion **2516** is inserted may be formed in the insulation unit included in the mating connector. The first-first connection protrusion **2516** may protrude from the first-first shield member **2511** along the vertical direction. The first-first connection protrusion **2516** and the first-first ground protrusion **2515** may protrude from the first-first shield member **2511** in directions opposite to each other with respect to the vertical direction. The first-first connection protrusion **2516** may be formed in a plate shape disposed in the vertical direction.

(66) The first-second ground contact **252** may be located between the first-second RF contact **211***b* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). Accordingly, the first-second ground contact **252** may shield between the first-second RF contact **211***b* and the transmit contacts **220**. The first-second ground contact **252** may be disposed to be spaced apart from the first-first ground contact **251** with respect to the second axial direction (Y-axis direction). The first-second ground contact **252** and the first-first ground contact **251** may be formed in different shapes. For example, the first-second ground contact **252** may be formed in a shape that does not have the first-first shield member **2511**, the first-first shield protrusion **2512**, the first-first ground protrusion **2515**, and the first-first connection protrusion **2516**, which are included in the first-first ground contact **251**. Accordingly, when compared with an embodiment in which the first-second ground contact **252** is formed in the same shape as the first-first ground contact **251**, in the board connector **200** according to the first embodiment, not only the easiness of a manufacturing operation may be improved in manufacturing the first-second ground contact **252**, but also material costs for manufacturing the first-second ground contact **252** may be reduced. In this case, shielding between the first-first RF contact **211***a* and the first-second RF contact **211***b* may be performed by the first-first ground contact **251**.

(67) The first-second ground contact **252** may include a first-second ground connection member **2521** and a first-second ground mounting member **2522**.

(68) The first-second ground connection member **2521** is provided to be connected to the ground contact included in the mating connector. Accordingly, the first ground contact **250** may be electrically connected to the ground contact included in the mating connector by being connected to the ground contact included in the mating connector through the first-second ground connection member **2521**. Accordingly, the gap generated as the first-second ground contact **252** and the first-first ground contact **251** are disposed to be spaced apart from each other along the second axial direction (Y-axis direction) may be shielded as the first ground contact **250** is connected to the

ground contact included in the mating connector through the first-second ground connection member **2521**. In this case, both the first-second ground connection member **2521** and the first-first ground connection member **2513** may be connected to the ground contact included in the mating connector.

(69) The first-second ground mounting member **2522** is mounted on the first board. The first-second ground mounting member **2522** may be grounded by being mounted on the first board. Accordingly, the first-second ground contact **252** may be grounded to the first board through the first-second ground mounting member **2522**. The first-second ground mounting member **2522** may protrude from the first-second ground connection member **2521** along the second axial direction (Y-axis direction). In this case, the first-second ground mounting member **2522** may be disposed between the first-second RF contact **211***b* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). The first-second ground mounting member **2522** may protrude from the first-second ground connection member **2521** by a length connectable to the ground housing **230** with respect to the second axial direction (Y-axis direction). In this case, the first-second ground mounting member **2522** and the first-first ground mounting member **2514** may protrude in opposite directions to be respectively connected to the sidewalls of the ground housing **230** facing each other. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding performance between the first RF contacts **211** and the transmit contact **220**. The first-second ground mounting member **2522** may be formed in a plate shape disposed in the horizontal direction.

(70) As described above, in the board connector **200** according to the first embodiment, a first ground loop **250***a* (illustrated in FIG. **5**) for the first-first RF contact **211***a* and the first-second RF contact **211***b* may be realized using the first-first ground contact **251**, the first-second ground contact **252**, and the ground housing **230**. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding performance for the first-first RF contact **211***a* and the first-second RF contact **211***b* using the first ground loop **250***a*, thereby realizing complete shielding for the first-first RF contact **211***a* and the first-second RF contact **211***b*.

(71) Referring to FIGS. **2** to **9**, when the plurality of second RF contacts **212** are provided, the second ground contact **260** may shield between the second RF contacts **212** and the transmit contacts **220** with respect to the first axial direction (X-axis direction), and also, shield between the second RF contacts **212** with respect to the second axial direction (Y-axis direction). Accordingly, by using the second ground contact **260**, the board connector **200** according to the first embodiment may realize a shielding function for between the second RF contacts **212** and the transmit contacts **220**, and also, additionally realize a shielding function for between the second RF contacts **212**. Accordingly, the board connector **200** according to the first embodiment may be realized to transmit a wider variety of RF signals using the second RF contacts **212**, thereby improving versatility applicable to a wider variety of electronic products.

(72) A second-first RF contact **212***a* among the second RF contacts **212** and a second-second RF contact **212***b* among the second RF contacts **212** may be coupled to the insulation unit **240** so as to be spaced apart from each other along the second axial direction (Y-axis direction). In FIG. **5**, the board connector **200** according to the first embodiment is illustrated as including two second RF contacts **212** realized as the second-first RF contact **212***a* and the second-second RF contact **212***b*, but the present disclosure is not limited thereto, and the board connector **200** according to the first embodiment may also include three or more second RF contacts **212**. Meanwhile, in the present specification, descriptions will be made on the basis of the case in which the board connector **200** according to the first embodiment includes the second-first RF contact **212***a* and the second-second RF contact **212***b*.

(73) When the second-first RF contact **212***a* and the second-second RF contact **212***b* are provided, the second ground contact **260** may include a second-first ground contact **261** and a second-second ground contact **262**.

(74) The second-first ground contact **261** may be located between the second-first RF contact **212***a* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). Accordingly, the second-first ground contact **261** may shield between the second-first RF contact **212***a* and the transmit contacts **220**.
(75) The second-first ground contact **261** may include a second-first shield member **2611**.
(76) The second-first shield member **2611** may be located between the second-first RF contact **212***a* and the second-second RF contact **212***b* with respect to the second axial direction (Y-axis direction). Accordingly, the second-first ground contact **261** may shield between the second-first RF contact **212***a* and the second-second RF contact **212***b* using the second-first shield member **2611**. Accordingly, even though the second-first RF contact **212***a* and the second-second RF contact **212***b* transmit different RF signals, the board connector **200** according to the first embodiment may prevent signals or the like from being interfered between the second-first RF contact **212***a* and the second-second RF contact **212***b* using the second-first shield member **2611**. Accordingly, the board connector **200** according to the first embodiment is realized to stably transmit a wider variety of RF signals using the second-first RF contact **212***a* and the second-second RF contact **212***b*. The second-first shield member **2611** may be formed in a plate shape disposed in the vertical direction between the second-first RF contact **212***a* and the second-second RF contact **212***b*.
(77) The second-first shield member **2611** may be disposed to be spaced apart from each of the second-first RF contact **212***a* and the second-second RF contact **212***b* with respect to the second axial direction (Y-axis direction) by the same distance. Accordingly, in the board connector **200** according to the first embodiment, a deviation between shielding performance for the second-first RF contact **212***a* and shielding performance for the second-second RF contact **212***b* may be reduced. Accordingly, the board connector **200** according to the first embodiment may stably realize a shielding function for each the second-first RF contact **212***a* and the second-second RF contact **212***b* using the second-first shield member **2611**.
(78) The second-first ground contact **261** may include a second-first shield protrusion **2612**.
(79) The second-first shield protrusion **2612** protrudes from the second-first shield member **2611**. The second-first shield protrusion **2612** may be connected to the ground housing **230**. Accordingly, the second ground contact **260** may enhance the shielding performance between the second-first RF contact **212***a* and the second-second RF contact **212***b* by being electrically connected to the ground housing **230** through the second-first shield protrusion **2612**, thereby realizing complete shielding. The second-first shield protrusion **2612** may be formed in a plate shape disposed in the vertical direction.
(80) The second-first ground contact **261** may include a second-first ground connection member **2613** and a second-first ground mounting member **2614**.
(81) The second-first ground connection member **2613** is coupled to each of the second-first shield member **2611** and the second-first ground mounting member **2614**. The second-first shield member **2611** and the second-first ground mounting member **2614** may be connected to each other through the second-first ground connection member **2613**. The second-first ground connection member **2613** may be connected to the ground contact included in the mating connector. Accordingly, the second ground contact **260** may be electrically connected to the ground contact included in the mating connector by being connected to the ground contact included in the mating connector through the second-first ground connection member **2613**. Accordingly, a gap generated as the second-first ground contact **261** and the second-second ground contact **262** are disposed to be spaced apart from each other along the second axial direction (Y-axis direction) may be shielded as the second ground contact **260** is connected to the ground contact included in the mating connector through the second-first ground connection member **2613**. The second-first shield member **2611** may be coupled to the second-first ground connection member **2613**. The second-first shield member **2611** may protrude from the second-first ground connection member **2613** along the first

axial direction (X-axis direction). In this case, the second-first shield protrusion **2612** may protrude from the second-first shield member **2611** along the first axial direction (X-axis direction).

(82) The second-first ground mounting member **2614** is mounted on the first board. The second-first ground mounting member **2614** may be grounded by being mounted on the first board. Accordingly, the second-first ground contact **261** may be grounded to the first board through the second-first ground mounting member **2614**. The second-first ground mounting member **2614** may protrude from the second-first ground connection member **2613** along the second axial direction (Y-axis direction). In this case, the second-first ground mounting member **2614** may be disposed between the second-first RF contact **212***a* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). The second-first ground mounting member **2614** may protrude from the second-first ground connection member **2613** by a length connectable to the ground housing **230** with respect to the second axial direction (Y-axis direction). In this case, the second-first ground mounting member **2614** and the second-first shield member **2611** protrude in different directions from the second-first ground connection member **2613** to be connected to different sidewalls included in the ground housing **230**. Accordingly, since the second-first ground contact **261** and the ground housing **230** are electrically connected to each other while surrounding all sides of the second-first RF contact **212***a*, the board connector **200** according to the first embodiment may further enhance the shielding performance for the second-first RF contact **212***a*, thereby realizing complete shielding. The second-first ground mounting member **2614** may be formed in a plate shape disposed in the horizontal direction.

(83) The second-first ground contact **261** may include a second-first ground protrusion **2615**.

(84) The second-first ground protrusion **2615** protrudes from the second-first shield member **2611**. The second-first ground protrusion **2615** may be mounted on the first board. Accordingly, a mounting area in which the second-first ground contact **261** is mounted on the first board may be increased, so that the board connector **200** according to the first embodiment may further enhance the shielding performance using the second-first ground contact **261**. The second-first ground protrusion **2615** may be mounted on the first board by passing through the insulation unit **240** and protruding from the insulation unit **240**. The second-first ground protrusion **2615** may protrude from the second-first shield member **2611** along the vertical direction. The second-first ground protrusion **2615** may be formed in a plate shape disposed in the vertical direction.

(85) The second-first ground contact **261** may include a second-first connection protrusion **2616**.

(86) The second-first connection protrusion **2616** protrudes from the second-first shield member **2611**. The second-first connection protrusion **2616** may be connected to the ground housing of the mating connector. Accordingly, a connection area in which the second-first ground contact **261** is connected to the ground housing of the mating connector may be increased, so that the board connector **200** according to the first embodiment may further enhance the shielding performance using the second-first ground contact **261**. The second-first connection protrusion **2616** may be connected to the ground housing of the mating connector by passing through the insulation unit **240** and protruding from the insulation unit **240**. The second-first connection protrusion **2616** may be inserted into the insulation unit included in the mating connector to be connected to the ground housing included in the mating connector. In this case, a through hole into which the second-first connection protrusion **2616** is inserted may be formed in the insulation unit included in the mating connector. The second-first connection protrusion **2616** may protrude from the second-first shield member **2611** along the vertical direction. The second-first connection protrusion **2616** and the second-first ground protrusion **2615** may protrude from the second-first shield member **2611** in directions opposite to each other with respect to the vertical direction. The second-first connection protrusion **2616** may be formed in a plate shape disposed in the vertical direction.

(87) The second-second ground contact **262** may be located between the second-second RF contact **212***b* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). Accordingly, the second-second ground contact **262** may shield between the second-second RF

contact **212***b* and the transmit contacts **220**. The second-second ground contact **262** may be disposed to be spaced apart from the second-first ground contact **261** with respect to the second axial direction (Y-axis direction). The second-second ground contact **262** and the second-first ground contact **261** may be formed in different shapes. For example, the second-second ground contact **262** may be formed in a shape that does not have the second-first shield member **2611**, the second-first shield protrusion **2612**, the second-first ground protrusion **2615**, and the second-first connection protrusion **2616**, which are included in the second-first ground contact **261**. Accordingly, when compared with an embodiment in which the second-second ground contact **262** is formed in the same shape as the second-first ground contact **261**, in the board connector **200** according to the first embodiment, not only the easiness of a manufacturing operation may be improved in manufacturing the second-second ground contact **262**, but also material costs for manufacturing the second-second ground contact **262** may be reduced. In this case, shielding between the second-first RF contact **212***a* and the second-second RF contact **212***b* may be performed by the second-first ground contact **261**.

(88) The second-second ground contact **262** may include a second-second ground connection member **2621** and a second-second ground mounting member **2622**.

(89) The second-second ground connection member **2621** is provided to be connected to the ground contact included in the mating connector. Accordingly, the second ground contact **260** may be electrically connected to the ground contact included in the mating connector by being connected to the ground contact included in the mating connector through the second-second ground connection member **2621**. Accordingly, the gap generated as the second-second ground contact **262** and the second-first ground contact **261** are disposed to be spaced apart from each other along the second axial direction (Y-axis direction) may be shielded as the second ground contact **260** is connected to the ground contact included in the mating connector through the second-second ground connection member **2621**. In this case, both the second-second ground connection member **2621** and the second-first ground connection member **2613** may be connected to the ground contact included in the mating connector.

(90) The second-second ground mounting member **2622** is mounted on the first board. The second-second ground mounting member **2622** may be grounded by being mounted on the first board. Accordingly, the second-second ground contact **262** may be grounded to the first board through the second-second ground mounting member **2622**. The second-second ground mounting member **2622** may protrude from the second-second ground connection member **2621** along the second axial direction (Y-axis direction). In this case, the second-second ground mounting member **2622** may be disposed between the second-second RF contact **212***b* and the transmit contacts **220** with respect to the first axial direction (X-axis direction). The second-second ground mounting member **2622** may protrude from the second-second ground connection member **2621** by a length connectable to the ground housing **230** with respect to the second axial direction (Y-axis direction). In this case, the second-second ground mounting member **2622** and the second-first ground mounting member **2614** may protrude in opposite directions to be respectively connected to the sidewalls of the ground housing **230** facing each other. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding performance between the second RF contacts **212** and the transmit contact **220**. The second-second ground mounting member **2622** may be formed in a plate shape disposed in the horizontal direction.

(91) As described above, in the board connector **200** according to the first embodiment, a second ground loop **260***a* (illustrated in FIG. **5**) for the second-first RF contact **212***a* and the second-second RF contact **212***b* may be realized using the second-first ground contact **261**, the second-second ground contact **262**, and the ground housing **230**. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding performance for the second-first RF contact **212***a* and the second-second RF contact **212***b* using the second ground loop **260***a*, thereby realizing complete shielding for the second-first RF contact **212***a* and the second-second

RF contact **212***b*.

(92) Here, the second-first ground contact **261** and the first-first ground contact **251** may be formed in the same shape. The second-second ground contact **262** and the first-second ground contact **252** may be formed in the same shape. Accordingly, in the board connector **200** according to the first embodiment, the easiness of a manufacturing operation may be improved in manufacturing each of the second-first ground contact **261**, the first-first ground contact **251**, the second-second ground contact **262**, and the first-second ground contact **252**.

(93) In this case, as shown in FIG. **5**, the second-first ground contact **261** and the first-first ground contact **251** may be disposed to be point-symmetric with respect to a symmetry point SP. The symmetry point SP is a point spaced apart from each of both sidewalls **230***b* and **230***c* of the ground housing **230**, which are disposed to be spaced apart from each other with respect to the first axial direction (X-axis direction), by the same distance, and also, spaced apart from each of both sidewalls **230***d* and **230***e* of the ground housing **230**, which are disposed to be spaced apart from each other with respect to the second axial direction (Y-axis direction), by the same distance. Accordingly, in the board connector **200** according to the first embodiment, the second-first ground contact **261** and the first-first ground contact **251** are formed in the same shape as each other and realized differently only in arrangement directions, and thus the easiness of a manufacturing operation may be further improved in manufacturing the second-first ground contact **261** and the first-first ground contact **251**. As shown in FIG. **5**, the second-second ground contact **262** and the first-second ground contact **252** may be disposed to be point-symmetric with respect to the symmetry point SP. Accordingly, in the board connector **200** according to the first embodiment, the second-second ground contact **262** and the first-second ground contact **252** are formed in the same shape and realized differently only in arrangement directions, and thus the easiness of a manufacturing operation may be further improved in manufacturing the second-second ground contact **262** and the first-second ground contact **252**. In this case, the second-first RF contact **212***a* and the first-first RF contact **211***a* may be disposed to be point-symmetric on the basis of the symmetry point SP. The second-second RF contact **212***b* and the first-second RF contact **211***b* may be disposed to be point-symmetric with respect to the symmetry point SP.

(94) Referring to FIGS. **2** to **10**, in the board connector **200** according to the first embodiment, the ground housing **230** may be realized as follows.

(95) The ground housing **230** may include a ground inner wall **231**, a ground outer wall **232**, and a ground connection wall **233**.

(96) The ground inner wall **231** faces the insulation unit **240**. The ground inner wall **231** may be disposed to face the inner side space **230***a*. The first-first ground contact **251** and the second-first ground contact **261** may each be connected to the ground inner wall **231**. The ground inner wall **231** may include a first sub-ground inner wall **2311**, a second sub-ground inner wall **2312**, a third sub-ground inner wall **2313**, and a fourth sub-ground inner wall **2314**.

(97) The first sub-ground inner wall **2311** and the second sub-ground inner wall **2312** may be disposed to face each other with respect to the first axial direction (X-axis direction). The third sub-ground inner wall **2313** and the fourth sub-ground inner wall **2314** may be disposed to face each other with respect to the second axial direction (Y-axis direction). The first sub-ground inner wall **2311**, the second sub-ground inner wall **2312**, the third sub-ground inner wall **2313**, and the fourth sub-ground inner wall **2314** may be coupled to the ground connection wall **233** at locations spaced apart from each other. Each of the first sub-ground inner wall **2311**, the second sub-ground inner wall **2312**, the third sub-ground inner wall **2313**, and the fourth sub-ground inner wall **2314** may elastically move with respect to a portion coupled to the ground connection wall **233** to press the insulation unit **240**. Accordingly, the board connector **200** according to the first embodiment may enhance a coupling force between the ground housing **230** and the insulation unit **240**. In addition, when the mating connector is inserted into the inner side space **230***a*, each of the first sub-ground inner wall **2311**, the second sub-ground inner wall **2312**, the third sub-ground inner wall **2313**, and

the fourth sub-ground inner wall **2314** may be pushed by the mating connector to more strongly press the insulation unit **240**, thereby further increasing the coupling force between the ground housing **230** and the insulation unit **240**.

(98) The ground outer wall **232** is spaced apart from the ground inner wall **231**. The ground outer wall **232** may be disposed outside the ground inner wall **231**. The ground outer wall **232** may be disposed to surround all sides of the ground inner wall **231**. The ground outer wall **232** and the ground inner wall **231** may be realized as shielding walls surrounding the sides of the inner side space **230***a*. The first RF contact **211** and the second RF contact **212** may be located in the inner side space **230***a* surrounded by the shielding walls. Accordingly, the ground housing **230** may realize a shielding function for the RF contacts **210** using the shielding walls. Accordingly, the board connector **200** according to the first embodiment may contribute to further improving the EMI shielding performance and the EMC performance using the shielding walls.

(99) The ground outer wall **232** may be grounded by being mounted on the first board. In this case, the ground housing **230** may be grounded through the ground outer wall **232**. When one end of the ground outer wall **232** is coupled to the ground connection wall **233**, the other end of the ground outer wall **232** may be mounted on the first board. In this case, the ground outer wall **232** may be formed at a higher height than the ground inner wall **231**.

(100) The ground outer wall **232** may be connected to the ground housing of the mating connector that is inserted into the inner side space **230***a*. For example, as shown in FIGS. **8** and **9**, the ground outer wall **232** may be connected to a ground housing **330** of the mating connector. As described above, the board connector **200** according to the first embodiment may further enhance the shielding function through the connection between the ground housing **230** and the ground housing of the mating connector. In addition, the board connector **200** according to the first embodiment may reduce electrical adverse effects such as crosstalk, which may be generated by mutual capacitance or induction between adjacent terminals due to the connection between the ground housing **230** and the ground housing of the mating connector. In this case, the board connector **200** according to the first embodiment may secure a path through which electromagnetic waves are introduced into at least one of the first board and the second board, thereby further enhancing the EMI shielding performance.

(101) The ground connection wall **233** is coupled to each of the ground inner wall **231** and the ground outer wall **232**. The ground connection wall **233** may be disposed between the ground inner wall **231** and the ground outer wall **232**. The ground inner wall **231** and the ground outer wall **232** may be electrically connected to each other through the ground connection wall **233**. Accordingly, when the ground outer wall **232** is mounted on the first board and grounded, the ground connection wall **233** and the ground inner wall **231** may also be grounded, thereby realizing a shielding function.

(102) The ground connection wall **233** may be coupled to one end of each of the ground outer wall **232** and the ground inner wall **231**. Based on FIG. **10**, one end of the ground outer wall **232** may correspond to an upper end of the ground outer wall **232**, and one end of the ground inner wall **231** may correspond to an upper end of the ground inner wall **231**. The ground connection wall **233** may be formed in a plate shape disposed in the horizontal direction, and the ground outer wall **232** and the ground inner wall **231** may each be formed in a plate shape disposed in the vertical direction. The ground connection wall **233**, the ground outer wall **232**, and the ground inner wall **231** may be integrally formed.

(103) The ground connection wall **233** may be connected to the ground housing of the mating connector that is inserted into the inner side space **230***a*. Accordingly, since the ground outer wall **232** and the ground connection wall **233** are connected to the ground housing of the mating connector, the board connector **200** according to the first embodiment may further enhance the shielding function by increasing a contact area between the ground housing **230** and the ground housing of the mating connector.

(104) Here, the ground housing **230** may realize a shielding function for the first RF contacts **211** together with the first ground contact **250**. The ground housing **230** may realize a shielding function for the second RF contacts **212** together with the second ground contact **260**.
(105) In this case, as shown in FIG. **5**, the ground housing **230** may include a first shielding wall **230***b*, a second shielding wall **230***c*, a third shielding wall **230***d*, and a fourth shielding wall **230***e*. Each of the first shielding wall **230***b*, the second shielding wall **230***c*, the third shielding wall **230***d*, and the fourth shielding wall **230***e* may be realized by the ground inner wall **231**, the ground outer wall **232**, and the ground connection wall **233**. The first shielding wall **230***b* and the second shielding wall **230***c* are disposed to face each other with respect to the first axial direction (X-axis direction). The first RF contacts **211** and the second RF contacts **212** may be located between the first shielding wall **230***b* and the second shielding wall **230***c* with respect to the first axial direction (X-axis direction). The first RF contacts **211** may be located at locations each having a shorter separation distance from the first shielding wall **230***b* than a separation distance from the second shielding wall **230***c* with respect to the first axial direction (X-axis direction). The second RF contacts **212** may be located at locations each having a shorter separation distance from the second shielding wall **230***c* than a separation distance from the first shielding wall **230***b* with respect to the first axial direction (X-axis direction). The third shielding wall **230***d* and the fourth shielding wall **230***e* are disposed to face each other with respect to the second axial direction (Y-axis direction). The first RF contacts **211** and the second RF contacts **212** may be located between the third shielding wall **230***d* and the fourth shielding wall **230***e* with respect to the second axial direction (Y-axis direction).
(106) The first ground contact **250** may be disposed between the first RF contacts **211** and the transmit contacts **220** with respect to the first axial direction (X-axis direction). Accordingly, the first RF contacts **211** may be located between the first shielding wall **230***b* and the first ground contact **250** with respect to the first axial direction (X-axis direction), and may be located between the third shielding wall **230***d* and the fourth shielding wall **230***e* with respect to the second axial direction (Y-axis direction). Thus, the board connector **200** according to the first embodiment may enhance the shielding function for the first RF contacts **211** using the first ground contact **250**, the first shielding wall **230***b*, the third shielding wall **230***d*, and the fourth shielding wall **230***e*. The first ground contact **250**, the first shielding wall **230***b*, the third shielding wall **230***d*, and the fourth shielding wall **230***e* are disposed at four sides of the first RF contacts **211** to realize a shielding force against RF signals. In this case, the first ground contact **250**, the first shielding wall **230***b*, the third shielding wall **230***d*, and the fourth shielding wall **230***e* may realize the first ground loop **250***a* (illustrated in FIG. **5**) for the first RF contacts **211**. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding function for the first RF contacts **211** using the first ground loop **250***a*, thereby realizing complete shielding for the first RF contacts **211**. In this case, the first RF contacts **211** may be located between the first sub-ground inner wall **2311** and the first ground contact **250** with respect to the first axial direction (X-axis direction), and also, located between the third sub-ground inner wall **2313** and the fourth sub-ground inner wall **2314** with respect to the second axial direction (Y-axis direction).
(107) The second ground contact **260** may be disposed between the second RF contacts **212** and the transmit contacts **220** with respect to the first axial direction (X-axis direction). Accordingly, the second RF contacts **212** may be located between the second shielding wall **230***c* and the second ground contact **260** with respect to the first axial direction (X-axis direction) and may be located between the third shielding wall **230***d* and the fourth shielding wall **230***e* with respect to the second axial direction (Y-axis direction). Accordingly, the board connector **200** according to the first embodiment may enhance the shielding function for the second RF contacts **212** using the second ground contact **260**, the second shielding wall **230***c*, the third shielding wall **230***d*, and the fourth shielding wall **230***e*. The second ground contact **260**, the second shielding wall **230***c*, the third shielding wall **230***d*, and the fourth shielding wall **230***e* are disposed at four sides of the second RF

contacts **212** to realize a shielding force against the RF signal. In this case, the second ground contact **260**, the second shielding wall **230***c*, the third shielding wall **230***d*, and the fourth shielding wall **230***e* may realize the second ground loop **260***a* (illustrated in FIG. **5**) for the second RF contacts **212**. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding function for the second RF contacts **212** using the second ground loop **260***a*, thereby realizing complete shielding for the second RF contacts **212**. In this case, the second RF contacts **212** may be located between the second sub-ground inner wall **2312** and the second ground contact **260** with respect to the first axial direction (X-axis direction), and also, located between the third sub-ground inner wall **2313** and the fourth sub-ground inner wall **2314** with respect to the second axial direction (Y-axis direction).
(108) The ground housing **230** may include a wedge member **234** (illustrated in FIG. **10**).
(109) The wedge member **234** protrudes from the ground inner wall **231**. When the ground housing **230** is coupled to the insulation unit **240**, the wedge member **234** may be wedged in the insulation unit **240** to fix the ground housing **230** and the insulation unit **240**. Accordingly, the board connector **200** according to the first embodiment may more firmly couple the ground housing **230** and the insulation unit **240** using the wedge member **234**. The wedge member **234** and the ground inner wall **231** may be integrally formed.
(110) The wedge member **234** may include a first wedge member **234***a* (illustrated in FIG. **8**) and a second wedge member **234***b* (illustrated in FIG. **8**).
(111) The first wedge member **234***a* protrudes from the first sub-ground inner wall **2311**. When the ground housing **230** is coupled to the insulation unit **240**, the first wedge member **234***a* may be wedged in the insulation unit **240** by being inserted into the insulation unit **240**, thereby fixing the ground housing **230** and the insulation unit **240**. The first-first ground contact **251** may be connected to the first wedge member **234***a*. In this case, the first-first shield protrusion **2512** may be electrically connected to the ground housing **230** by being connected to the first wedge member **234***a*. Accordingly, the first wedge member **234***a* may enhance the coupling force between the ground housing **230** and the insulation unit **240**, and simultaneously, enhance the shielding performance between the first-first RF contact **211***a* and the first-second RF contact **211***b*.
(112) The second wedge member **234***b* protrudes from the second sub-ground inner wall **2312**. When the ground housing **230** is coupled to the insulation unit **240**, the second wedge member **234***b* may be wedged in the insulation unit **240** by being inserted into the insulation unit **240**, thereby fixing the ground housing **230** and the insulation unit **240**. The second wedge member **234***b* and the first wedge member **234***a* may be disposed to face each other with respect to the first axial direction (X-axis direction). The second wedge member **234***b* may be connected to the second-first ground contact **261**. In this case, the second-first shield protrusion **2612** may be electrically connected to the ground housing **230** by being connected to the second wedge member **234***b*. Accordingly, the second wedge member **234***b* may enhance the coupling force between the ground housing **230** and the insulation unit **240**, and simultaneously, enhance the shielding performance between the second-first RF contact **212***a* and the second-second RF contact **212***b*.
(113) Referring to FIGS. **2** to **14**, the ground housing **230** may include the following configuration in order to further enhance the shielding function by improving a contact between the ground inner wall **231** and the ground housing of the mating connector.
(114) First, as shown in FIG. **11**, the ground housing **230** may include a connection groove **235**. The connection groove **235** may be formed on an outer surface of the ground outer wall **232**. The outer surface of the ground outer wall **232** is a surface facing a side opposite to the inner side space **230***a*. The connection groove **235** may be realized as a groove formed to a predetermined depth in the outer surface of the ground outer wall **232**. The ground housing **330** included in the mating connector may be inserted into the connection groove **235**. In this case, a connection protrusion **336** included in the ground housing **330** of the mating connector may be inserted into the connection groove **235**. Accordingly, the board connector **200** according to the first embodiment may further

enhance the shielding function for the first RF contact **211** and the second RF contact **212** by improving a contact between the ground housing **230** and the ground housing **330** included in the mating connector using the connection groove **235**. In FIG. **11**, the connection groove **235** is illustrated as being formed to have a longer length than the connection protrusion **336** with respect to the vertical direction, but the present disclosure is not limited thereto, and the connection groove **235** and the connection protrusion **336** may be formed to have lengths substantially equal to each other. Meanwhile, the ground outer wall **232** may support the connection protrusion **336** that is inserted into the connection groove **235**, so that the connection protrusion **336** is prevented from being separated from the connection groove **235**. The ground housing **230** may also include a plurality of connection grooves **235**. In this case, the connection grooves **235** may be disposed to be spaced apart from each other along the outer surface of the ground outer wall **232**.

(115) Next, as shown in FIG. **12**, the ground housing **230** may also include a connection protrusion **236**. The connection protrusion **236** may be formed on the outer surface of the ground outer wall **232**. The connection protrusion **236** may protrude from the outer surface of the ground outer wall **232**. The connection protrusion **236** may be inserted into the ground housing **330** included in the mating connector. In this case, the connection protrusion **236** may be inserted into a connection groove **337** included in the ground housing **330** of the mating connector. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding function for the first RF contact **211** and the second RF contact **212** by improving the contact between the ground housing **230** and the ground housing **330** included in the mating connector using the connection protrusion **236**. In FIG. **12**, the connection protrusion **236** is illustrated as being formed to have a shorter length than the connection groove **337** with respect to the vertical direction, but the present disclosure is not limited thereto, and the connection protrusion **236** and the connection groove **337** may be formed to have lengths substantially equal to each other. Meanwhile, the connection protrusion **236** is inserted into the connection groove **337** to be supported by the ground housing **330**, so that the connection protrusion **236** may be prevented from being separated from the connection groove **337**. The ground housing **230** may also include a plurality of connection protrusions **236**. In this case, the connection protrusions **236** may be disposed to be spaced apart from each other along the outer surface of the ground outer wall **232**.

(116) Next, as shown in FIG. **13**, when the ground housing **230** includes the connection protrusion **236**, the connection protrusion **236** may be supported by the connection protrusion **336** included in the ground housing **330** of the mating connector. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding function for the first RF contact **211** and the second RF contact **212** by improving the contact between the ground housing **230** and the ground housing **330** included in the mating connector using the connection protrusion **236**. Meanwhile, the connection protrusion **236** may be disposed on a lower side of the connection protrusion **336** to be supported by the connection protrusion **336**, so that the connection protrusion **236** may also be prevented from being separated from the connection protrusion **336**.

(117) Next, as shown in FIG. **8**, the ground housing **230** may be in contact with the ground housing **330** of the mating connector as the outer surface of the ground outer wall **232** is brought into surface contact with the ground housing **330** of the mating connector. In this case, a gap may occur between the outer surface of the ground outer wall **232** and the ground housing **330** of the mating connector, and in order to compensate for the gap, as shown in FIG. **14**, the ground housing **230** may include a conductive member **237**. The conductive member **237** may be coupled to the outer surface of the ground outer wall **232**. The conductive member **237** may extend along the outer surface of the ground outer wall **232**, including a corner portion **232***a* (illustrated in FIG. **10**) included in the outer surface of the ground outer wall **232**, to form a closed loop shape. Accordingly, the board connector **200** according to the first embodiment may further enhance the shielding function for the first RF contact **211** and the second RF contact **212** by improving the contact between the ground housing **230** and the ground housing **330** included in the mating

connector using the conductive member **237**. In addition, in the case of the embodiment using the connection protrusion **236** and the connection groove **235**, it is difficult to realize the connection protrusion **236** and the connection groove **235** at the corner portion **232***a* included in the outer surface of the ground outer wall **232**, but in the case of the embodiment using the conductive member **237**, it is possible to improve the easiness of the operation of realizing the conductive member **237** at the corner portion **232***a* included in the outer surface of the ground outer wall **232**. The conductive member **237** may be formed of an electrically conductive material to electrically connect the ground outer wall **232** and the ground housing **330** of the mating connector. For example, the conductive member **237** may be formed of metal. After the conductive member **237** is separately manufactured, the conductive member **237** may be coupled to the ground outer wall **232** by being mounted, attached, fastened, and the like to the outer surface of the ground outer wall **232**. The conductive member **237** may also be coupled to the ground outer wall **232** by applying a conductive shielding material to the outer surface of the ground outer wall **232**.

(118) Referring to FIG. **15**, the ground housing **230** may be realized as double shielding walls. In this case, the ground inner wall **231** may be disposed to surround all sides of the inner side space **230***a*. Accordingly, the ground housing **230** may be realized as double shielding walls in which the ground inner wall **231** and the ground outer wall **232** are disposed to surround all sides of the inner side space **230***a*. Accordingly, the ground housing **230** may enhance the shielding function for the RF contacts **210** using the double shielding walls. Accordingly, the board connector **200** according to the first embodiment may contribute to further improving the EMI shielding performance and the EMC performance using the double shielding walls.

(119) Referring to FIGS. **2** to **16**, in the board connector **200** according to the first embodiment, the insulation unit **240** may be realized as follows.

(120) The insulation unit **240** may include an insulating member **241**, an insertion member **242**, and a connecting member **243**.

(121) The insulating member **241** supports the RF contacts **210** and the transmit contacts **220**. The insulating member **241** may be located in the inner side space **230***a*. The insulating member **241** may be located inside the ground inner wall **231**. The insulating member **241** may be inserted into an inner side space included in the mating connector.

(122) The insertion member **242** is inserted between the ground inner wall **231** and the ground outer wall **232**. As the insertion member **242** is inserted between the ground inner wall **231** and the ground outer wall **232**, the insulation unit **240** may be coupled to the ground housing **230**. The insertion member **242** may be inserted between the ground inner wall **231** and the ground outer wall **232** in an interference fit manner. The insertion member **242** may be disposed outside the insulating member **241**. The insertion member **242** may be disposed to surround the outside of the insulating member **241**.

(123) The connecting member **243** is coupled to each of the insertion member **242** and the insulating member **241**. The insertion member **242** and the insulating member **241** may be connected to each other through the connecting member **243**. The connecting member **243** may be formed to have a thickness less than that of each of the insertion member **242** and the insulating member **241** with respect to the vertical direction. Accordingly, a space may be provided between the insertion member **242** and the insulating member **241**, and the mating connector may be inserted into the space. The connecting member **243**, the insertion member **242**, and the connecting member **243** may be integrally formed.

(124) The insulation unit **240** may include a soldering inspection window **244** (illustrated in FIG. **7**).

(125) The soldering inspection window **244** may be formed by passing through the insulation unit **240**. The soldering inspection window **244** may be used to inspect a state in which the first RF mounting members **2111** are mounted on the first board. In this case, the first RF contacts **211** may be coupled to the insulation unit **240** such that the first RF mounting members **2111** are located in

the soldering inspection windows **244**. Accordingly, the first RF mounting members **2111** are not covered by the insulation unit **240**. Accordingly, in a state in which the board connector **200** according to the first embodiment is mounted on the first board, a worker may inspect the state, in which first RF mounting members **2111** is mounted on the first board, through the soldering inspection window **244**. Accordingly, in the board connector **200** according to the first embodiment, even when all of the first RF contacts **211** including the first RF mounting members **2111** are located inside the ground housing **230**, the accuracy of a mounting operation of mounting the first RF contacts **211** on the first board may be improved. The soldering inspection window **244** may be formed by passing through the insulating member **241**.
(126) The insulation unit **240** may also include a plurality of soldering inspection windows **244**. In this case, the first RF mounting members **2111** may be located in different soldering inspection windows **244**, respectively. The second RF mounting members **2121** and the transmission mounting members **2201** may be located in the soldering inspection windows **244**, respectively. Accordingly, in the state in which the board connector **200** according to the first embodiment is mounted on the first board, a worker may inspect the state, in which the first RF mounting members **2111**, the second RF mounting members **2121**, and the transmission mounting members **2201** are mounted on the first board, through the soldering inspection windows **244**. Accordingly, the board connector **200** according to the first embodiment may improve the accuracy of the operation of mounting the first RF contacts **211**, the second RF contacts **212**, and the transmit contacts **220** on the first board. The soldering inspection windows **244** may be formed by passing through the insulation unit **240** at locations spaced apart from each other.
(127) The insulation unit **240** may include a first assembly groove **245** (illustrated in FIG. **16**).
(128) The first wedge member **234***a* (illustrated in FIG. **8**) is inserted into the first assembly groove **245**. As the first wedge member **234***a* is inserted into the first assembly groove **245**, the first wedge member **234***a* may be wedged in the insulation unit **240** to fix the ground housing **230** and the insulation unit **240**. The first assembly groove **245** may be realized as a groove formed to a predetermined depth in the insulating member **241**. The first-first shield protrusion **2512** may be inserted into the first assembly groove **245**. The first-first shield protrusion **2512** may be inserted into the first assembly groove **245** to be connected to the first wedge member **234***a*. Accordingly, the first-first ground contact **251** may be electrically connected to the ground housing **230**.
(129) The insulation unit **240** may include a second assembly groove **246** (illustrated in FIG. **16**).
(130) The second wedge member **234***b* (illustrated in FIG. **8**) is inserted into the second assembly groove **246**. As the second wedge member **234***b* is inserted into the second assembly groove **246**, the second wedge member **234***b* may be wedged in the insulation unit **240** to fix the ground housing **230** and the insulation unit **240**. The second assembly groove **246** may be realized as a groove formed to a predetermined depth in the insulating member **241**. The second-first shield protrusion **2612** may be inserted into the second assembly groove **246**. The second-first shield protrusion **2612** may be inserted into the second assembly groove **246** to be connected to the second wedge member **234***b*. Accordingly, the second-first ground contact **261** may be electrically connected to the ground housing **230**.
(131) <Board Connector **300** According to Second Embodiment>
(132) Referring to FIGS. **2**, **17**, and **18**, the board connector **300** according to the second embodiment may include a plurality of RF contacts **310**, a plurality of transmit contacts **320**, a ground housing **330**, and an insulation unit **340**.
(133) The RF contacts **310** are for transmitting RF signals. The RF contacts **310** may transmit ultra-high frequency RF signals. The RF contacts **310** may be supported by the insulation unit **340**. The RF contacts **310** may be coupled to the insulation unit **340** through an assembly process. The RF contacts **310** may also be integrally molded with the insulation unit **340** through injection molding.
(134) The RF contacts **310** may be disposed to be spaced apart from each other. The RF contacts **310** may be electrically connected to the second board by being mounted on the second board. The

RF contacts **310** may be electrically connected to the first board, on which a mating connector is mounted, by being connected to RF contacts included in the mating connector. Accordingly, the second board and the first board may be electrically connected. In this case, the mating connector may be realized as the board connector **200** according to the first embodiment. Meanwhile, the mating connector in the board connector **200** according to the first embodiment may also be realized as the board connector **300** according to the second embodiment.

(135) A first RF contact **311** among the RF contacts **310** and a second RF contact **312** among the RF contacts **310** may be spaced apart from each other along the first axial direction (X-axis direction). The first RF contact **311** and the second RF contact **312** may be supported by the insulation unit **340** at locations spaced apart from each other along the first axial direction (X-axis direction).

(136) The first RF contact **311** may include a first RF mounting member **3111**. The first RF mounting member **3111** may be mounted on the second board. Accordingly, the first RF contact **311** may be electrically connected to the second board through the first RF mounting member **3111**. The first RF contact **311** may be formed of an electrically conductive material. For example, the first RF contact **311** may be formed of metal. The first RF contact **311** may be connected to any one of the RF contacts included in the mating connector.

(137) The second RF contact **312** may include a second RF mounting member **3121**. The second RF mounting member **3121** may be mounted on the second board. Accordingly, the second RF contact **312** may be electrically connected to the second board through the second RF mounting member **3121**. The second RF contact **312** may be formed of an electrically conductive material. For example, the second RF contact **312** may be formed of metal. The second RF contact **312** may be connected to any one of the RF contacts included in the mating connector.

(138) Referring to FIGS. **2**, **16**, and **17**, the transmit contacts **320** are coupled to the insulation unit **340**. The transmit contacts **320** may serve to transmit signals, data, and the like. The transmit contacts **320** may be coupled to the insulation unit **340** through an assembly process. The transmit contacts **320** may also be integrally molded with the insulation unit **340** through injection molding.

(139) The transmit contacts **320** may be disposed between the first RF contact **311** and the second RF contact **312** with respect to the first axial direction (X-axis direction). Accordingly, in order to reduce RF signal interference between the first RF contact **311** and the second RF contact **312**, the transmit contacts **320** may be disposed in a space in which the first RF contact **311** and the second RF contact **312** are spaced apart. Accordingly, the board connector **300** according to the second embodiment may not only reduce RF signal interference by increasing a distance by which the first RF contact **311** and the second RF contact **312** are spaced apart from each other, but also improve space utilization for the insulation unit **340** by disposing the transmit contacts **320** in a separation space for this purpose.

(140) The transmit contacts **320** may be disposed to be spaced apart from each other. The transmit contacts **320** may be electrically connected to the second board by being mounted on the second board. In this case, a transmission mounting member **3201** included in each of the transmit contacts **320** may be mounted on the second board. The transmit contacts **320** may be formed of an electrically conductive material. For example, the transmit contacts **320** may be formed of metal. The transmit contacts **320** may be electrically connected to the second board, on which the mating connector is mounted, by being connected to transmit contacts included in the mating connector. Accordingly, the second board and the first board may be electrically connected.

(141) Meanwhile, in FIG. **18**, the board connector **300** according to the second embodiment is illustrated as including four transmit contacts **320**, but the present disclosure is not limited thereto, and the board connector **300** according to the second embodiment may also include five or more transmit contacts **320**. The transmit contacts **320** may be spaced apart from each other along the first axial direction (X-axis direction) and the second axial direction (Y-axis direction).

(142) Referring to FIGS. **17** to **19**, the ground housing **330** is coupled to the insulation unit **340**.

The ground housing **330** may be grounded by being mounted on the second board. Accordingly, the ground housing **330** may realize a shielding function against signals, electromagnetic waves, or the like for the RF contacts **310**. In this case, the ground housing **330** may prevent electromagnetic waves generated from the RF contacts **310** from interfering with signals of circuit components located around the electronic device, and may prevent electromagnetic waves generated from the circuit components located around the electronic device from interfering with RF signals transmitted by the RF contacts **310**. Accordingly, the board connector **300** according to the second embodiment may contribute to improving EMI shielding performance and EMC performance using the ground housing **330**. The ground housing **330** may be formed of an electrically conductive material. For example, the ground housing **330** may be formed of metal.
(143) The ground housing **330** may be disposed to surround sides of an inner side space **330***a*. The insulation unit **340** may be located in the inner side space **330***a*. All of the first RF contact **311**, the second RF contact **312**, and the transmit contacts **220** may be located in the inner side space **330***a*. In this case, all of the first RF mounting member **3111**, the second RF mounting member **3121**, and the transmission mounting members **3201** may also be located in the inner side space **330***a*. Accordingly, the ground housing **330** may enhance a shielding function for the first RF contact **311** and the second RF contact **312** by realizing shielding walls for all of the first RF contact **311** and the second RF contact **312**, thereby realizing complete shielding. The mating connector may be inserted into the inner side space **330***a*. In this case, a portion of the mating connector may be inserted into the inner side space **330***a*, and a portion of the board connector **300** according to the second embodiment may be inserted into an inner side space included in the mating connector.
(144) The ground housing **330** may be disposed to surround all sides of the inner side space **330***a*. The inner side space **330***a* may be disposed inside the ground housing **330**. When the entire ground housing **330** is formed in a rectangular loop shape, the inner side space **330***a* may be formed in a rectangular parallelepiped shape. In this case, the ground housing **330** may be disposed to surround four sides of the inner side space **330***a*.
(145) The ground housing **330** may be integrally formed as one piece without a seam. The ground housing **330** may be integrally formed as one piece without a seam by a metal injection method, such as a metal die casting method, an MIM method, or the like. The ground housing **330** may be integrally formed as one piece without a seam by a CNC process, an MCT process, or the like.
(146) Referring to FIGS. **17** to **19**, the insulation unit **340** supports the RF contacts **310**. The RF contacts **310** and the transmit contacts **320** may be coupled to the insulation unit **340**. The insulation unit **340** may be formed of an insulating material. The insulation unit **340** may be coupled to the ground housing **330** such that the RF contacts **310** are located in the inner side space **330***a*.
(147) Referring to FIGS. **9** and **17** to **20**, the board connector **300** according to the second embodiment may include a first ground contact **350**.
(148) The first ground contact **350** is coupled to the insulation unit **340**. The first ground contact **350** may be grounded by being mounted on the second board. The first ground contact **350** may be coupled to the insulation unit **340** through an assembly process. The first ground contact **350** may also be integrally molded with the insulation unit **340** through injection molding.
(149) The first ground contact **350** may realize a shielding function for the first RF contact **311** together with the ground housing **330**. In this case, the first ground contact **350** may be disposed between the first RF contact **311** and the transmit contacts **320** with respect to the first axial direction (X-axis direction). The first ground contact **350** may be formed of an electrically conductive material. For example, the first ground contact **350** may be formed of metal. When the mating connector is inserted into the inner side space **330***a*, the first ground contact **350** may be connected to a ground contact included in the mating connector.
(150) Although not illustrated in the drawings, the board connector **300** according to the second embodiment may also include a plurality of first ground contacts **350**. The first ground contacts **350**

may be disposed to be spaced apart from each other along the second axial direction (Y-axis direction). A gap, which is formed as the first ground contacts **350** are spaced apart from each other, may be blocked as the first ground contact **350** is connected to the ground contact included in the mating connector.
(151) Referring to FIGS. **9** and **17** to **20**, the board connector **300** according to the second embodiment may include a second ground contact **360**.
(152) The second ground contact **360** is coupled to the insulation unit **340**. The second ground contact **360** may be grounded by being mounted on the second board. The second ground contact **360** may be coupled to the insulation unit **340** through an assembly process. The second ground contact **360** may also be integrally molded with the insulation unit **340** through injection molding.
(153) The second ground contact **360** may realize a shielding function for the second RF contact **312** together with the ground housing **330**. The second ground contact **360** may be disposed between the transmit contacts **320** and the second RF contact **312** with respect to the first axial direction (X-axis direction). The second ground contact **360** may be formed of an electrically conductive material. For example, the second ground contact **360** may be formed of metal. When the mating connector is inserted into the inner side space **330***a*, the second ground contact **360** may be connected to the ground contact included in the mating connector.
(154) Although not illustrated in the drawings, the board connector **300** according to the second embodiment may also include a plurality of second ground contact **360**. The second ground contacts **360** may be disposed to be spaced apart from each other along the second axial direction (Y-axis direction). A gap, which is formed as the second ground contacts **360** are spaced apart from each other, may be blocked as the second ground contact **360** is connected to the ground contact included in the mating connector.
(155) Here, the board connector **300** according to the second embodiment may be realized to include a plurality of first RF contacts **311** and a plurality of second RF contacts **312**.
(156) Referring to FIGS. **9** and **17** to **21**, the first RF contacts **311** and the second RF contacts **312** may be disposed to be spaced apart from each other along the first axial direction (X-axis direction). The transmit contacts **320** may be disposed between the first RF contacts **311** and the second RF contacts **312** with respect to the first axial direction (X-axis direction). In this case, the first ground contact **350** may shield between the first RF contacts **311** and the transmit contacts **320** with respect to the first axial direction (X-axis direction). The second ground contact **360** may shield between the second RF contacts **312** and the transmit contacts **320** with respect to the first axial direction (X-axis direction).
(157) When the plurality of first RF contacts **311** are provided, the first ground contact **350** may shield between the first RF contacts **311** and the transmit contacts **320** with respect to the first axial direction (X-axis direction). As the first ground contact **350** is connected to the ground contact included in the mating connector, between the first RF contacts **311** with respect to the second axial direction (Y-axis direction) may be shielded. Accordingly, by using the first ground contact **350**, the board connector **300** according to the second embodiment may realize a shielding function for between the first RF contacts **311** and the transmit contacts **320**, and also, additionally realize a shielding function for between the first RF contacts **311** using the connection between the first ground contact **350** and the ground contact included in the mating connector. In this case, the board connector **300** according to the second embodiment may shield between the first RF contacts **311** using the ground housing **330**. Accordingly, the board connector **300** according to the second embodiment may be realized to transmit a wider variety of RF signals using the first RF contacts **311**, thereby improving versatility applicable to a wider variety of electronic products.
(158) A first-first RF contact **311***a* among the first RF contacts **311** and a first-second RF contact **311***b* among the first RF contacts **311** may be coupled to the insulation unit **340** so as to be spaced apart from each other along the second axial direction (Y-axis direction). In FIG. **21**, the board connector **300** according to the second embodiment is illustrated as including two first RF contacts

311 realized as the first-first RF contact **311***a* and the first-second RF contact **311***b*, but the present disclosure is not limited thereto, and the board connector **300** according to the second embodiment may also include three or more first RF contacts **311**. Meanwhile, in the present specification, descriptions will be made on the basis of the case in which the board connector **300** according to the second embodiment includes the first-first RF contact **311***a* and the first-second RF contact **311***b*.
(159) When the first-first RF contact **311***a* and the first-second RF contact **311***b* are provided, the first ground contact **350** may include a first ground mounting member **351** (illustrated in FIG. **9**) and a first ground connection member **352** (illustrated in FIG. **9**).
(160) The first ground mounting member **351** is mounted on the second board. The first ground mounting member **351** may be grounded by being mounted on the second board. Accordingly, the first ground contact **350** may be grounded to the second board through the first ground mounting member **351**. The first ground mounting member **351** may be disposed along the second axial direction (Y-axis direction). In this case, the first ground mounting member **351** may be disposed between the first-first RF contact **311***a* and the transmit contacts **320** with respect to the first axial direction (X-axis direction). The first ground mounting member **351** may also be disposed between the first-second RF contact **311***b* and the transmit contacts **320** with respect to the first axial direction (X-axis direction). The first ground mounting member **351** may be formed in a plate shape disposed in the vertical direction. The first ground mounting member **351** may be connected to the ground contact included in the mating connector. For example, as shown in FIG. **8**, the first ground mounting member **351** may be connected to the first-first ground connection member **2513** included in the mating connector.
(161) The first ground connection member **352** is coupled to the first ground mounting member **351**. The first ground connection member **352** may protrude from the first ground mounting member **351** along the vertical direction. The first ground connection member **352** may be connected to the ground contact included in the mating connector. Accordingly, the first ground contact **350** may be electrically connected to the ground contact included in the mating connector by being connected to the ground contact included in the mating connector through the first ground connection member **352**. Accordingly, the first ground contact **350** may realize a shielding force for the first-first RF contact **311***a* and the first-second RF contact **311***b* through the connection between the first ground connection member **352** and the ground contact included in the mating connector. In this case, the first ground contact **350** may realize a shielding force that shields between each of the first-first RF contact **311***a* and the first-second RF contact **311***b* and the transmit contacts **320** with respect to the first axial direction (X-axis direction). In this case, the first ground contact **350** may realize a shielding force that shields between the first-first RF contact **311***a* and the first-second RF contact **311***b* with respect to the second axial direction (Y-axis direction). The first ground connection member **352** may be formed in a plate shape disposed in the vertical direction.
(162) The first ground contact **350** may include a plurality of first ground connection members **352**. First ground connection members **352** and **352′** (illustrated in FIG. **9**) may be disposed to be spaced apart from each other along the second axial direction (Y-axis direction). The first ground connection members **352** may be connected to different ground contacts included in the mating connectors, respectively. For example, as shown in FIG. **9**, the first ground connection members **352** and **352′** may be respectively connected to the first-first ground contact **251** and the first-second ground contact **252** included in the mating connector. In this case, the first ground connection member **352** may be connected to the first-first ground connection member **2513** included in the first-first ground contact **251**. The first ground connection member **352′** may be connected to the first-second ground connection member **2521** included in the first-second ground contact **252**. When the mating connector is inserted into the inner side space **330***a*, the first-first shield member **2511** of the first-first ground contact **251** included in the mating connector may be located between the first-first RF contact **311***a* and the first-second RF contact **311***b* with respect to

the second axial direction (Y-axis direction).
(163) As described above, the board connector **300** according to the second embodiment may realize a first ground loop **350***a* (illustrated in FIG. **21**) for the first-first RF contact **311***a* and the first-second RF contact **311***b* using the connection between the first ground contact **350** and the ground contact included in the mating connector. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding performance for the first-first RF contact **311***a* and the first-second RF contact **311***b* using the first ground loop **350***a*, thereby realizing complete shielding for the first-first RF contact **311***a* and the first-second RF contact **311***b*.
(164) When the plurality of second RF contacts **312** are provided, the second ground contact **360** may shield between the second RF contacts **312** and the transmit contacts **320** with respect to the first axial direction (X-axis direction). As the second ground contact **360** is connected to the ground contact included in the mating connector, shielding may be provided between the second RF contacts **312** with respect to the second axial direction (Y-axis direction). Accordingly, by using the second ground contact **360**, the board connector **300** according to the second embodiment may realize a shielding function for between the second RF contacts **312** and the transmit contacts **320**, and also, additionally realize a shielding function for between the second RF contacts **312** using the connection between the second ground contact **360** and the ground contact included in the mating connector. In this case, the board connector **300** according to the second embodiment may shield between the second RF contacts **312** using the ground housing **330**. Accordingly, the board connector **300** according to the second embodiment may be realized to transmit a wider variety of RF signals using the second RF contacts **312**, thereby improving versatility applicable to a wider variety of electronic products.
(165) A second-first RF contact **312***a* among the second RF contacts **312** and a second-second RF contact **312***b* among the second RF contacts **312** may be coupled to the insulation unit **340** so as to be spaced apart from each other along the second axial direction (Y-axis direction). In FIG. **21**, the board connector **300** according to the second embodiment is illustrated as including two second RF contacts **312** realized as the second-first RF contact **312***a* and the second-second RF contact **312***b*, but the present disclosure is not limited thereto, and the board connector **300** according to the second embodiment may also include three or more second RF contacts **312**. Meanwhile, in the present specification, descriptions will be made on the basis of the case in which the board connector **300** according to the second embodiment includes the second-first RF contact **312***a* and the second-second RF contact **312***b*.
(166) When the second-first RF contact **312***a* and the second-second RF contact **312***b* are provided, the second ground contact **360** may include a second ground mounting member **361** (illustrated in FIG. **20**) and a second ground connection member **362** (illustrated in FIG. **20**).
(167) The second ground mounting member **361** is mounted on the second board. The second ground mounting member **361** may be grounded by being mounted on the second board. Accordingly, the second ground contact **360** may be grounded to the second board through the second ground mounting member **361**. The second ground mounting member **361** may be disposed along the second axial direction (Y-axis direction). In this case, the second ground mounting member **361** may be disposed between the second-first RF contact **312***a* and the transmit contacts **320** with respect to the first axial direction (X-axis direction). The second ground mounting member **361** may also be disposed between the second-second RF contact **312***b* and the transmit contacts **320** with respect to the first axial direction (X-axis direction). The second ground mounting member **361** may be formed in a plate shape disposed in the vertical direction. The second ground mounting member **361** may be connected to the ground contact included in the mating connector. For example, as shown in FIG. **8**, the second ground mounting member **361** may be connected to the second-first ground connection member **2613** included in the mating connector.
(168) The second ground connection member **362** is coupled to the second ground mounting

member **361**. The second ground connection member **362** may protrude from the second ground mounting member **361** along the vertical direction. The second ground connection member **362** may be connected to the ground contact included in the mating connector. Accordingly, the second ground contact **360** may be electrically connected to the ground contact included in the mating connector by being connected to the ground contact included in the mating connector through the second ground connection member **362**. Accordingly, the second ground contact **360** may realize a shielding force for the second-first RF contact **312***a* and the second-second RF contact **312***b* through the connection between the second ground connection member **362** and the ground contact included in the mating connector. In this case, the second ground contact **360** may realize a shielding force that shields between each of the second-first RF contact **312***a* and the second-second RF contact **312***b* and the transmit contacts **320** with respect to the first axial direction (X-axis direction). In this case, the second ground contact **360** may realize a shielding force that shields between the second-first RF contact **312***a* and the second-second RF contact **312***b* with respect to the second axial direction (Y-axis direction). The second ground connection member **362** may be formed in a plate shape disposed in the vertical direction.

(169) The second ground contact **360** may include a plurality of second ground connection members **362**. Second ground connection members **362** and **362**′ (illustrated in FIG. **20**) may be disposed to be spaced apart from each other along the second axial direction (Y-axis direction). The second ground connection members **362** may be connected to different ground contacts included in the mating connectors, respectively. For example, as shown in FIG. **20**, the second ground connection members **362** and **362**′ may be respectively connected to the second-first ground contact **261** and the second-second ground contact **262** included in the mating connector. In this case, the second ground connection member **362** may be connected to the second-first ground connection member **2613** included in the second-first ground contact **261**. The second ground connection member **362**′ may be connected to the second-second ground connection member **2621** included in the second-second ground contact **262**. When the mating connector is inserted into the inner side space **330***a*, the second-first shield member **2611** of the second-first ground contact **261** included in the mating connector may be located between the second-first RF contact **312***a* and the second-second RF contact **312***b* with respect to the second axial direction (Y-axis direction).

(170) As described above, the board connector **300** according to the second embodiment may realize a second ground loop **360***a* (illustrated in FIG. **21**) for the second-first RF contact **312***a* and the second-second RF contact **312***b* using the connection between the second ground contact **360** and the ground contact included in the mating connector. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding performance for the second-first RF contact **312***a* and the second-second RF contact **312***b* using the second ground loop **360***a*, thereby realizing complete shielding for the second-first RF contact **312***a* and the second-second RF contact **312***b*.

(171) Referring to FIGS. **8**, **9**, and **11** to **23**, in the board connector **300** according to the second embodiment, the ground housing **330** may be realized as follows.

(172) The ground housing **330** may include a ground sidewall **331** and a ground bottom **332**.

(173) The ground sidewall **331** is disposed to surround a side of the inner side space **330***a*. The ground sidewall **331** may be disposed to surround all sides of the inner side space **330***a*. When the mating connector is inserted into the inner side space **330***a*, the ground sidewall **331** may be connected to the ground housing included in the mating connector. For example, the ground sidewall **331** may be connected to the ground outer wall **232** of the ground housing **230** included in the mating connector. The ground sidewall **331** may be formed in a plate shape disposed in the vertical direction.

(174) The ground bottom **332** protrudes from a lower end of the ground sidewall **331** toward the inner side space **330***a*. That is, the ground bottom **332** may protrude to the inside of the ground sidewall **331**. The ground bottom **332** may extend along the lower end of the ground sidewall **331**

and formed in a closed loop shape. The ground bottom **332** may be grounded by being mounted on the second board. Accordingly, the ground sidewall **331** may be grounded through the ground bottom **332**. In this case, the ground housing **330** may be grounded through the ground bottom **332**. When the mating connector is inserted into the inner side space **330***a*, the ground bottom **332** may be connected to the ground housing included in the mating connector. For example, the ground bottom **332** may be connected to the ground connection wall **233** of the ground housing **230** included in the mating connector. The ground bottom **332** may be formed in a plate shape disposed in the horizontal direction.

(175) The ground bottom **332** and the ground sidewall **331** may be disposed to surround the inner side space **330***a*. In this case, the first RF contact **311** and the second RF contact **312** may be located in the inner side space **330***a* surrounded by the ground bottom **332** and the ground sidewall **331**. Accordingly, the ground bottom **332** and the ground sidewall **331** may enhance the shielding function for the first RF contact **311** and the second RF contact **312** by realizing shielding walls for all of the first RF contact **311** and the second RF contact **312**, thereby realizing complete shielding.

(176) The ground bottom **332** and the ground sidewall **331** may be integrally formed. In this case, the ground housing **330** may be integrally formed as one piece without a seam. The ground housing **330** may be integrally formed as one piece without a seam by a metal injection method, such as a metal die casting method, an MIM method, or the like. The ground housing **330** may be integrally formed as one piece without a seam by a CNC process, an MCT process, or the like.

(177) The ground housing **330** may include a first shield bottom **333**.

(178) The first shield bottom **333** protrudes from the ground bottom **332**. The first shield bottom **333** protrudes from the ground bottom **332** toward the first ground contact **350**, and thus may be located between the first-first RF contact **311***a* and the first-second RF contact **311***b* with respect to the second axial direction (Y-axis direction). Accordingly, the first shield bottom **333** may shield between the first-first RF contact **311***a* and the first-second RF contact **311***b* with respect to the second axial direction (Y-axis direction). The first shield bottom **333** may be formed in a plate shape disposed in the vertical direction.

(179) The first shield bottom **333** may be connected to the ground contact included in the mating connector. For example, as shown in FIG. **8**, the first shield bottom **333** may be connected to the first-first ground contact **251** included in the mating connector. In this case, the first shield bottom **333** may be connected to the first-first connection protrusion **2516** included in the first ground contact **250**. Accordingly, the board connector **300** according to the second embodiment may realize the first ground loop **350***a* for the first-first RF contact **311***a* and the first-second RF contact **311***b* using the connection between the first shield bottom **333** and the ground contact included in the mating connector. The first shield bottom **333** and the ground bottom **332** may be integrally formed.

(180) The ground housing **330** may include a second shield bottom **334**.

(181) The second shield bottom **334** protrudes from the ground bottom **332**. The second shield bottom **334** protrudes from the ground bottom **332** toward the second ground contact **360**, and thus may be located between the second-first RF contact **312***a* and the second-second RF contact **312***b* with respect to the second axial direction (Y-axis direction). Accordingly, the second shield bottom **334** may shield between the second-first RF contact **312***a* and the second-second RF contact **312***b* with respect to the second axial direction (Y-axis direction). The second shield bottom **334** may be formed in a plate shape disposed in the vertical direction.

(182) The second shield bottom **334** may be connected to the ground contact included in the mating connector. For example, as shown in FIG. **8**, the second shield bottom **334** may be connected to the second-first ground contact **261** included in the mating connector. In this case, the second shield bottom **334** may be connected to the second-first connection protrusion **2616** included in the second-first ground contact **261**. Accordingly, the board connector **300** according to the second embodiment may realize the second ground loop **360***a* for the second-first RF contact **312***a* and the

second-second RF contact **312***b* using the connection between the second shield bottom **334** and the ground contact included in the mating connector. The second shield bottom **334** and the ground bottom **332** may be integrally formed.
(183) The ground housing **330** may include a ground top wall **335**.
(184) The ground top wall **335** protrudes from an upper end of the ground sidewall **331** to a side opposite to the inner side space **330***a*. In this case, the ground top wall **335** may protrude toward the outside of the ground sidewall **331**. The ground top wall **335** may extend along the upper end of the ground sidewall **331** and may be formed in a closed loop shape. The ground top wall **335** may be formed in a plate shape disposed in the horizontal direction.
(185) The ground top wall **335**, the ground bottom **332**, and the ground sidewall **331** may be integrally formed. In this case, the ground housing **330** may be integrally formed as one piece without a seam. The ground housing **330** may be integrally formed as one piece without a seam by a metal injection method, such as a metal die casting method, an MIM method, or the like. The ground housing **330** may be integrally formed as one piece without a seam by a CNC process, an MCT process, or the like.
(186) A connection portion of the ground top wall **335** and the ground sidewall **331** may be formed to be rounded as shown in FIGS. **8** and **9**. Accordingly, the connection portion between the ground top wall **335** and the ground sidewall **331** may serve as a guide for the mating connector when the mating connector is inserted into the inner side space **330***a*. In this case, in the connection portion of the ground top wall **335** and the ground sidewall **331**, a portion facing the inner side space **330***a* may be formed to be rounded.
(187) The ground top wall **335**, the ground sidewall **331**, and the ground bottom **332** may realize shielding walls. In this case, as shown in FIGS. **19** and **21**, the ground housing **330** may include a first shielding wall **330***b*, a second shielding wall **330***c*, a third shielding wall **330***d*, and a fourth shielding wall **330***e*. Each of the first shielding wall **330***b*, the second shielding wall **330***c*, the third shielding wall **330***d*, and the fourth shielding wall **330***e* may be realized by the ground sidewall **331**, the ground bottom **332**, and the ground top wall **335**. The first shielding wall **330***b* and the second shielding wall **330***c* are disposed to face each other with respect to the first axial direction (X-axis direction). The first RF contacts **311** and the second RF contacts **312** may be located between the first shielding wall **330***b* and the second shielding wall **330***c* with respect to the first axial direction (X-axis direction). The first RF contacts **311** may be located at locations each having a shorter separation distance from the first shielding wall **330***b* than a separation distance from the second shielding wall **330***c* with respect to the first axial direction (X-axis direction). The second RF contacts **312** may be located at locations each having a shorter separation distance from the second shielding wall **330***c* than a separation distance from the first shielding wall **330***b* with respect to the first axial direction (X-axis direction). The third shielding wall **330***d* and the fourth shielding wall **330***e* are disposed to face each other with respect to the second axial direction (Y-axis direction). The first RF contacts **311** and the second RF contacts **312** may be located between the third shielding wall **330***d* and the fourth shielding wall **330***e* with respect to the second axial direction (Y-axis direction).
(188) In this case, the first ground contact **350**, the first shielding wall **330***b*, the third shielding wall **330***d*, the fourth shielding wall **330***e*, and the first shield bottom **333** may realize the first ground loop **350***a* (illustrated in FIG. **21**) for the first-first RF contact **311***a* and the first-second RF contact **311***b*. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding function for the first-first RF contact **311***a* and the first-second RF contact **311***b* using the first ground loop **350***a*, thereby realizing complete shielding for the first-first RF contact **311***a* and the first-second RF contact **311***b*.
(189) In this case, the second ground contact **360**, the second shielding wall **330***c*, the third shielding wall **330***d*, the fourth shielding wall **330***e*, and the second shield bottom **334** may realize the second ground loop **360***a* (illustrated in FIG. **21**) for the second-first RF contact **312***a* and the

second-second RF contact **312***b*. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding function for the second-first RF contact **312***a* and the second-second RF contact **312***b* using the second ground loop **360***a*, thereby realizing complete shielding for the second-first RF contact **312***a* and the second-second RF contact **312***b*.

(190) Referring to FIGS. **8** to **13** and **23**, the ground housing **330** may include the following configuration in order to further enhance the shielding function by improving the contact between the ground sidewall **331** and the ground housing of the mating connector.

(191) First, as shown in FIG. **11**, the ground housing **330** may include the connection protrusion **336**. The connection protrusion **336** may be formed on an inner surface of the ground sidewall **331**. The connection protrusion **336** may protrude from the inner surface of the ground sidewall **331**. The connection protrusion **336** may be inserted into the ground housing **230** included in the mating connector. In this case, the connection protrusion **336** may be inserted into the connection groove **235** included in the ground housing **230** of the mating connector. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding function for the first RF contact **311** and the second RF contact **312** by improving the contact between the ground housing **330** and the ground housing **230** included in the mating connector using the connection protrusion **336**. In FIG. **11**, the connection protrusion **336** is illustrated as being formed to have a shorter length than the connection groove **235** with respect to the vertical direction, but the present disclosure is not limited thereto, and the connection protrusion **336** and the connection groove **235** may be formed to have lengths substantially equal to each other. The ground housing **330** may also include a plurality of connection protrusions **336**. In this case, the connection protrusions **336** may be disposed to be spaced apart from each other along the inner surface of the ground sidewall **331**.

(192) Next, as shown in FIG. **12**, the ground housing **330** may include the connection groove **337**. The connection groove **337** may be formed on the inner surface of the ground sidewall **331**. The connection groove **337** may be realized as a groove formed to a predetermined depth in the inner surface of the ground sidewall **331**. The ground housing **230** included in the mating connector may be inserted into the connection groove **337**. In this case, the connection protrusion **236** included in the ground housing **230** of the mating connector may be inserted into the connection groove **337**. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding function for the first RF contact **311** and the second RF contact **312** by improving the contact between the ground housing **330** and the ground housing **230** included in the mating connector using the connection groove **337**. In FIG. **10**, the connection groove **337** is illustrated as being formed to have a longer length than the connection protrusion **236** with respect to the vertical direction, but the present disclosure is not limited thereto, and the connection groove **337** and the connection protrusion **236** may be formed to have lengths substantially equal to each other. Meanwhile, the ground sidewall **331** may support the connection protrusion **236** that is inserted into the connection groove **337**, so that the connection protrusion **236** is prevented from being separated from the connection groove **337**. The ground housing **330** may also include a plurality of connection grooves **337**. In this case, the connection grooves **337** may be disposed to be spaced apart from each other along the inner surface of the ground sidewall **331**.

(193) Next, as shown in FIG. **13**, when the ground housing **330** includes the connection protrusion **336**, the connection protrusion **336** may be supported by the connection protrusion **236** included in the ground housing **230** of the mating connector. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding function for the first RF contact **311** and the second RF contact **312** by improving the contact between the ground housing **330** and the ground housing **230** included in the mating connector using the connection protrusion **336**. Meanwhile, the connection protrusion **336** may be disposed on an upper side of the connection protrusion **236** to support the connection protrusion **236**.

(194) Next, as shown in FIG. **8**, the ground housing **330** may be in contact with the ground housing **330** of the mating connector as the inner surface of the ground sidewall **331** is brought into surface

contact with the ground housing **330** of the mating connector. In this case, a gap may occur between the inner surface of the ground sidewall **331** and the ground housing **230** of the mating connector, and in order to compensate for the gap, as shown in FIG. **23**, the ground housing **330** may include a conductive member **338**. The conductive member **338** may be coupled to the inner surface of the ground sidewall **331**. The conductive member **338** may extend along the inner surface of the ground sidewall **331**, including a corner portion **3301** (illustrated in FIG. **22**) included in the inner surface of the ground sidewall **331** to form a closed loop shape. Accordingly, the board connector **300** according to the second embodiment may further enhance the shielding function for the first RF contact **311** and the second RF contact **312** by improving the contact between the ground housing **330** and the ground housing **230** included in the mating connector using the conductive member **338**. In addition, in the case of the embodiment using the connection protrusion **336** and the connection groove **337**, it is difficult to realize the connection protrusion **336** and the connection groove **337** at the corner portion **3301** included in the inner surface of the ground sidewall **331**, but in the case of the embodiment using the conductive member **338**, it is possible to improve the easiness of the operation of realizing the conductive member **338** at the corner portion **3301** included in the inner surface of the ground sidewall **331**. The conductive member **338** may be formed of an electrically conductive material to electrically connect the ground sidewall **331** and the ground housing **230** of the mating connector. For example, the conductive member **338** may be formed of metal. After the conductive member **338** is separately manufactured, the conductive member **338** may be coupled to the ground sidewall **331** by being mounted, attached, fastened, and the like to the inner surface of the ground sidewall **331**. The conductive member **338** may also be coupled to the ground sidewall **331** by applying a conductive shielding material to the inner surface of the ground sidewall **331**.

(195) Referring to FIGS. **17** to **23**, the ground housing **330** may include a coupling member **339**.

(196) The coupling member **339** protrudes upward from the ground bottom **332**. When the ground housing **330** is coupled to the insulation unit **340**, the coupling member **339** may be inserted into the insulation unit **340**. Accordingly, the coupling member **339** may firmly couple the ground housing **330** and the insulation unit **340**. The coupling member **339** may also be coupled to the insulation unit **340** in an interference fit manner. The coupling member **339** and the ground bottom **332** may be integrally formed. A coupling groove (not shown) for inserting the coupling member **339** thereto may be formed in the insulation unit **340**. The coupling groove may be formed on a lower surface of the insulation unit **340**.

(197) The ground housing **330** may also include a plurality of coupling members **339**. In this case, the coupling members **339** may be disposed to be spaced apart from each other along the ground bottom **332**. In FIG. **22**, the ground housing **330** is illustrated as including four coupling members **339**, but the present disclosure is not limited thereto, and the ground housing **330** may also include two, three, or five or more coupling members **339**. The coupling grooves may be formed in the insulation unit **340** in the same number as the coupling members **339**.

(198) The ground housing **330** may include a wedge member **3391** protruding from the coupling member **339**. As the coupling member **339** is inserted into the insulation unit **340**, the wedge member **3391** may be wedged in the insulation unit **340** to fix the ground housing **330** and the insulation unit **340**. Accordingly, the board connector **300** according to the second embodiment may more firmly couple the ground housing **330** and the insulation unit **340** using the wedge member **3391**. When the coupling member **339** is disposed to be spaced apart from the ground sidewall **331** along the second axial direction (Y-axis direction), the wedge member **3391** may protrude from a side surface of the coupling member **339** along the first axial direction (X-axis direction). The wedge member **3391** and the coupling member **339** may be integrally formed.

(199) Referring to FIGS. **17** to **23**, in the board connector **300** according to the second embodiment, the insulation unit **340** may include a soldering inspection window **341** (illustrated in FIG. **19**).

(200) The soldering inspection window **341** may be formed by passing through the insulation unit

340. The soldering inspection window **341** may be used to inspect a state in which the first RF mounting members **3111** are mounted on the second board. In this case, the first RF contacts **311** may be coupled to the insulation unit **340** such that the first RF mounting members **3111** are located in the soldering inspection window **341**. Accordingly, the first RF mounting members **3111** are not covered by the insulation unit **340**. Accordingly, in a state in which the board connector **300** according to the second embodiment is mounted on the second board, a worker may inspect the state, in which first RF mounting members **3111** are mounted on the second board, through the soldering inspection window **341**. Accordingly, in the board connector **300** according to the second embodiment, even when all of the first RF contacts **311** including the first RF mounting members **3111** are located inside the ground housing **330**, the accuracy of a mounting operation of mounting the first RF contacts **311** on the second board may be improved. The soldering inspection window **341** may be formed by passing through the insulating member **241**.
(201) The insulation unit **340** may also include a plurality of soldering inspection windows **341**. In this case, the first RF mounting members **3111** may be located in different soldering inspection windows **341**, respectively. The second RF mounting members **3121** and the transmission mounting members **3201** may be located in the soldering inspection windows **341**, respectively. Accordingly, in the state in which the board connector **300** according to the second embodiment is mounted on the second board, a worker may inspect the state, in which the first RF mounting members **3111**, the second RF mounting members **3121**, and the transmission mounting members **3201** are mounted on the second board, through the soldering inspection windows **341**. Accordingly, the board connector **300** according to the second embodiment may improve the accuracy of the operation of mounting the first RF contacts **311**, the second RF contacts **312**, and the transmit contacts **320** on the second board. The soldering inspection windows **341** may be formed by passing through the insulation unit **340** at locations spaced apart from each other.
(202) Hereinafter, an embodiment of a mounting pattern of the board, on which the board connector according to the present disclosure is mounted, will be described in detail with reference to the accompanying drawings.
(203) FIGS. **24** to **27** are conceptual bottom views illustrating an embodiment of a mounting pattern of a board on which the board connector according to the first embodiment is mounted, and FIGS. **28** to **31** are conceptual bottom views illustrating an embodiment of a mounting pattern of a board on which the board connector according to the second embodiment is mounted. FIGS. **24** to **27** illustrate a location of the mounting pattern on the basis of a bottom surface of the board connector according to the first embodiment described with reference to FIG. **5**. FIGS. **28** to **31** illustrate a location of the mounting pattern on the basis of a bottom surface of the board connector according to the second embodiment described with reference to FIG. **21**. In FIGS. **24** to **31**, hatched regions are the locations of the mounting patterns.
(204) Referring to FIGS. **24** to **27**, the board connector **200** according to the first embodiment may be mounted on a mounting pattern **201** formed on the board (not shown). As the board connector **200** according to the first embodiment is electrically connected to the mounting pattern **201**, a shielding force for the RF contacts **210** may be enhanced. The board connector **200** according to the first embodiment may be mounted on the mounting pattern **201** realized in various embodiments, and the embodiments of the mounting pattern **201** will be described sequentially with reference to the accompanying drawings.
(205) First, as shown in FIG. **24**, the mounting pattern **201** may be formed on the board in a shape surrounding the inner side space **230***a*. For example, the mounting pattern **201** may be formed in a rectangular loop shape along an outer side of the inner side space **230***a*. The ground housing **230** may be mounted on the mounting pattern **201**. When the ground housing **230** is mounted on the mounting pattern **201**, the shielding force for the RF contacts **210** may be enhanced through an electrical connection between the ground housing **230** and the mounting pattern **201**. In this case, the shielding force by the mounting pattern **201** may be realized in a form surrounding all of the

contacts located in the inner side space **230***a*.
(206) Next, as shown in FIG. **25**, a first mounting pattern **201***a*, a second mounting pattern **201***b*, a third mounting pattern **201***c*, and a fourth mounting pattern **201***d* may be formed on the board. The first mounting pattern **201***a*, the second mounting pattern **201***b*, the third mounting pattern **201***c*, and the fourth mounting pattern **201***d* may be disposed to be spaced apart from each other. The ground housing **230** may be mounted on each of the first mounting pattern **201***a*, the second mounting pattern **201***b*, the third mounting pattern **201***c*, and the fourth mounting pattern **201***d*. In this case, different shielding walls **230***b*, **230***c*, **230***d*, and **230***e* included in the ground housing **230** may be mounted on the first mounting pattern **201***a*, the second mounting pattern **201***b*, the third mounting pattern **201***c*, and the fourth mounting pattern **201***d*, respectively. Accordingly, the shielding force for the RF contacts **210** may be enhanced through electrical connections between the ground housing **230** and the mounting patterns **201***a*, **201***b*, **201***c*, and **201***d*.
(207) Next, as shown in FIG. **26**, the first mounting pattern **201***a* and the second mounting pattern **201***b* may be formed on the board. The first mounting pattern **201***a* and the second mounting pattern **201***b* may be disposed to be spaced apart from each other along the first axial direction (X-axis direction).
(208) The first ground contact **250** may be mounted on the first mounting pattern **201***a*. Accordingly, a shielding force for the first RF contact **211** may be enhanced through an electrical connection between the first mounting pattern **201***a* and the first ground contact **250**. In this case, a portion of the first-first ground contact **251** and all of the first-second ground contact **252** may be mounted on the first mounting pattern **201***a*. The first ground contact **250** and the ground housing **230** may also be mounted on the first mounting pattern **201***a*. In this case, the third shielding wall **230***d* and the fourth shielding wall **230***e* may be mounted on the first mounting pattern **201***a*. Thus, the shielding force for the first RF contact **211** may be further enhanced. The first mounting pattern **201***a* may be formed to extend parallel to the second axial direction (Y-axis direction).
(209) The second ground contact **260** may be mounted on the second mounting pattern **201***b*. Accordingly, a shielding force for the second RF contact **212** may be enhanced through an electrical connection between the second mounting pattern **201***b* and the second ground contact **260**. In this case, a portion of the second-first ground contact **261** and all of the second-second ground contact **262** may be mounted on the second mounting pattern **201***b*. The second ground contact **260** and the ground housing **230** may also be mounted on the second mounting pattern **201***b*. In this case, the third shielding wall **230***d* and the fourth shielding wall **230***e* may be mounted on the second mounting pattern **201***b*. Thus, the shielding force for the second RF contact **212** may be further enhanced. The second mounting pattern **201***b* may be formed to extend parallel to the second axial direction (Y-axis direction).
(210) Next, as shown in FIG. **27**, the first mounting pattern **201***a* and the second mounting pattern **201***b* may be formed on the board. The first mounting pattern **201***a* and the second mounting pattern **201***b* may be disposed to be spaced apart from each other along the first axial direction (X-axis direction).
(211) The first ground contact **250** may be mounted on the first mounting pattern **201***a*. All of the first-first ground contact **251** and all of the first-second ground contact **252** may be mounted on the first mounting pattern **201***a*. Accordingly, a shielding force between the first RF contact **211** and the second RF contact **212** may be enhanced through the electrical connection between the first mounting pattern **201***a* and the first ground contact **250**, and a shielding force between the first-first RF contact **211***a* and the first-second RF contact **211***b* may also be enhanced. The first ground contact **250** and the ground housing **230** may be mounted on the first mounting pattern **201***a*. In this case, the first shielding wall **230***b*, the third shielding wall **230***d*, and the fourth shielding wall **230***e* may be mounted on the first mounting pattern **201***a*. Accordingly, the shielding force between the first RF contact **211** and the second RF contact **212** and the shielding force between the first-first RF contact **211***a* and the first-second RF contact **211***b* may be further enhanced. The first

mounting pattern **201***a* may be formed in a shape in which a portion extending parallel to the second axial direction (Y-axis direction) and a portion extending parallel to the first axial direction (X-axis direction) are combined. For example, the first mounting pattern **201***a* may be formed in a T-shape as a whole.

(212) The second ground contact **260** may be mounted on the second mounting pattern **201***b*. All of the second-first ground contact **261** and all of the second-second ground contact **262** may be mounted on the second mounting pattern **201***b*. Accordingly, a shielding force between the second RF contact **212** and the first RF contact **211** may be enhanced through the electrical connection between the second mounting pattern **201***b* and the second ground contact **260**, and a shielding force between the second-first RF contact **212***a* and the second-second RF contact **212***b* may also be enhanced. The second ground contact **260** and the ground housing **230** may also be mounted on the second mounting pattern **201***b*. In this case, the second shielding wall **230***c*, the third shielding wall **230***d*, and the fourth shielding wall **230***e* may be mounted on the second mounting pattern **201***b*. Accordingly, the shielding force between the first RF contact **211** and the second RF contact **212** and the shielding force between the second-first RF contact **212***a* and the second-second RF contact **212***b* may be further enhanced. The second mounting pattern **201***b* may be formed in a shape in which a portion extending parallel to the second axial direction (Y-axis direction) and a portion extending parallel to the first axial direction (X-axis direction) are combined. For example, the second mounting pattern **201***b* may be formed in a T-shape as a whole. The second mounting pattern **201***b* and the first mounting pattern **201***a* may be formed in a shape symmetrical to each other.

(213) Referring to FIGS. **28** to **31**, the board connector **300** according to the second embodiment may be mounted on a mounting pattern **301** formed on the board (not shown). As the board connector **300** according to the second embodiment is electrically connected to the mounting pattern **301**, the shielding force for the RF contacts **310** may be enhanced. The board connector **300** according to the second embodiment may be mounted on the mounting pattern **301** realized in various embodiments, and the embodiments of the mounting pattern **301** will be described sequentially with reference to the accompanying drawings.

(214) First, as shown in FIG. **28**, the mounting pattern **301** may be formed on the board in a shape surrounding the inner side space **330***a*. For example, the mounting pattern **301** may be formed in a rectangular loop shape along an outer side of the inner side space **330***a*. The ground housing **330** may be mounted on the mounting pattern **301**. When the ground housing **330** is mounted on the mounting pattern **301**, a shielding force for the RF contacts **310** may be enhanced through an electrical connection between the ground housing **330** and the mounting pattern **301**. In this case, the shielding force by the mounting pattern **301** may be realized in a form surrounding all of the contacts located in the inner side space **330***a*.

(215) Next, as shown in FIG. **29**, a first mounting pattern **301***a*, a second mounting pattern **301***b*, a third mounting pattern **301***c*, and a fourth mounting pattern **301***d* may be formed on the board. The first mounting pattern **301***a*, the second mounting pattern **301***b*, the third mounting pattern **301***c*, and the fourth mounting pattern **301***d* may be disposed to be spaced apart from each other. The ground housing **330** may be mounted on each of the first mounting pattern **301***a*, the second mounting pattern **301***b*, the third mounting pattern **301***c*, and the fourth mounting pattern **301***d*. In this case, different shielding walls **330***b*, **330***c*, **330***d*, and **330***e* included in the ground housing **330** may be mounted on the first mounting pattern **301***a*, the second mounting pattern **301***b*, the third mounting pattern **301***c*, and the fourth mounting pattern **301***d*, respectively. Accordingly, the shielding force for the RF contacts **310** may be enhanced through electrical connections between the ground housing **330** and the mounting patterns **301***a*, **301***b*, **301***c*, and **301***d*.

(216) Next, as shown in FIG. **30**, the first mounting pattern **301***a* and the second mounting pattern **301***b* may be formed on the board. The first mounting pattern **301***a* and the second mounting pattern **301***b* may be disposed to be spaced apart from each other along the first axial direction (X-

axis direction).

(217) The first ground contact **350** may be mounted on the first mounting pattern **301***a*. Accordingly, a shielding force for the first RF contact **311** may be enhanced through an electrical connection between the first mounting pattern **301***a* and the first ground contact **350**. In this case, all of the first ground contact **350** and a portion of the first shield bottom **333** may be mounted on the first mounting pattern **301***a*. The first ground contact **350** and the ground housing **330** may also be mounted on the first mounting pattern **301***a*. In this case, the third shielding wall **330***d* and the fourth shielding wall **330***e* may be mounted on the first mounting pattern **301***a*. Thus, the shielding force for the first RF contact **311** may be further enhanced. The first mounting pattern **301***a* may be formed to extend parallel to the second axial direction (Y-axis direction).

(218) The second ground contact **360** may be mounted on the second mounting pattern **301***b*. Accordingly, a shielding force for the second RF contact **312** may be enhanced through an electrical connection between the second mounting pattern **301***b* and the second ground contact **360**. In this case, all of the second ground contact **360** and a portion of the second shield bottom **334** may be mounted on the second mounting pattern **301***b*. The second ground contact **360** and the ground housing **330** may also be mounted on the second mounting pattern **301***b*. In this case, the third shielding wall **330***d* and the fourth shielding wall **330***e* may be mounted on the second mounting pattern **301***b*. Thus, the shielding force for the second RF contact **312** may be further enhanced. The second mounting pattern **301***b* may be formed to extend parallel to the second axial direction (Y-axis direction).

(219) Next, as shown in FIG. **31**, the first mounting pattern **301***a* and the second mounting pattern **301***b* may be formed on the board. The first mounting pattern **301***a* and the second mounting pattern **301***b* may be disposed to be spaced apart from each other along the first axial direction (X-axis direction).

(220) The first ground contact **350** may be mounted on the first mounting pattern **301***a*. All of the first ground contact **350** and all of the first shield bottom **333** may be mounted on the first mounting pattern **301***a*. Accordingly, a shielding force between the first RF contact **311** and the second RF contact **312** may be enhanced through the electrical connection between the first mounting pattern **301***a* and the first ground contact **350**, and a shielding force between the first-first RF contact **311***a* and the first-second RF contact **311***b* may be enhanced through the electrical connection between the first mounting pattern **301***a* and the first shield bottom **333**. The first ground contact **350** and the ground housing **330** may also be mounted on the first mounting pattern **301***a*. In this case, the third shielding wall **330***d* and the fourth shielding wall **330***e* may be mounted on the first mounting pattern **301***a*. Thus, the shielding force between the first RF contact **311** and the second RF contact **312** and the shielding force between the first-first RF contact **311***a* and the first-second RF contact **311***b* may be further enhanced. Although not shown in the drawings, the first shielding wall **330***b*, the third shielding wall **330***d*, and the fourth shielding wall **330***e* may also be mounted on the first mounting pattern **301***a*. The first mounting pattern **301***a* may be formed in a shape in which a portion extending parallel to the second axial direction (Y-axis direction) and a portion extending parallel to the first axial direction (X-axis direction) are combined. For example, the first mounting pattern **301***a* may be formed in a T-shape as a whole.

(221) The second ground contact **360** may be mounted on the second mounting pattern **301***b*. All of the second ground contact **360** and all of the second shield bottom **334** may be mounted on the second mounting pattern **301***b*. Accordingly, a shielding force between the second RF contact **312** and the first RF contact **311** may be enhanced through the electrical connection between the second mounting pattern **301***b* and the second ground contact **360**, and a shielding force between the second-first RF contact **312***a* and the second-second RF contact **312***b* may be enhanced through an electrical connection between the second mounting pattern **301***b* and the second shield bottom **334**. The second ground contact **360** and the ground housing **330** may also be mounted on the second mounting pattern **301***b*. In this case, the third shielding wall **330***d* and the fourth shielding wall

330*e* may be mounted on the second mounting pattern **301***b*. Accordingly, the shielding force between the second RF contact **312** and the first RF contact **311** and the shielding force between the second-first RF contact **312***a* and the second-second RF contact **312***b* may be further enhanced. Although not shown in the drawings, the second shielding wall **330***c*, the third shielding wall **330***d*, and the fourth shielding wall **330***e* may also be mounted on the second mounting pattern **301***b*. The second mounting pattern **301***b* may be formed in a shape in which a portion extending parallel to the second axial direction (Y-axis direction) and a portion extending parallel to the first axial direction (X-axis direction) are combined. For example, the second mounting pattern **301***b* may be formed in a T-shape as a whole. The second mounting pattern **301***b* and the first mounting pattern **301***a* may be formed in a shape symmetrical to each other.
(222) It should be understood that the present disclosure is not limited to the above-described embodiments and the accompanying drawings, and various substitutions, modifications, and alterations can be devised by those skilled in the art to which the present disclosure pertains without departing from the technical spirit of the embodiments described herein.

Claims

1. A board connector comprising: a plurality of radio frequency (RF) contacts for transmitting an RF signal; an insulation unit configured to support the RF contacts; a plurality of transmit contacts that are coupled to the insulation unit between a plurality of first RF contacts among the RF contacts and a plurality of second RF contacts among the RF contacts such that the first RF contacts and the second RF contacts are spaced apart from each other along a first axial direction; a ground housing to which the insulation unit is coupled; a first ground contact coupled to the insulation unit and configured to shield between the first RF contacts and the transmit contacts with respect to the first axial direction; and a second ground contact coupled to the insulation unit and configured to shield between the second RF contacts and the transmit contacts with respect to the first axial direction, wherein the first ground contact shields between the first RF contacts and the transmit contacts with respect to the first axial direction, and shields between the first RF contacts with respect to a second axial direction perpendicular to the first axial direction.
2. The board connector of claim 1, wherein the first ground contact includes a first-first ground contact located between a first-first RF contact among the first RF contacts and the transmit contacts with respect to the first axial direction, and a first-second ground contact located between a first-second RF contact among the first RF contacts and the transmit contacts with respect to the first axial direction, wherein the first-first ground contact includes a first-first shield member located between the first-first RF contact and the first-second RF contact with respect to the second axial direction.
3. The board connector of claim 2, wherein the first-first ground contact includes a first-first shield protrusion protruding from the first-first shield member, wherein the first-first shield protrusion is connected to the ground housing.
4. The board connector of claim 3, wherein the ground housing includes a first sub-ground inner wall facing the insulation unit and a first wedge member protruding from the first sub-ground inner wall, and the first-first shield protrusion is electrically connected to the ground housing by being connected to the first wedge member.
5. The board connector of claim 2, wherein the first-first ground contact includes a first-first ground mounting member mounted on a board, and a first-first ground connection member coupled to each of the first-first ground mounting member and the first-first shield member, the first-first shield member protrudes from the first-first ground connection member along the first axial direction, and the first-first ground mounting member protrudes from the first-first ground connection member along the second axial direction.
6. The board connector of claim 2, wherein the first-first ground contact includes a first-first

ground protrusion protruding from the first-first shield member, wherein the first-first ground protrusion is mounted on a board.
7. The board connector of claim 2, wherein the first-first ground contact includes a first-first connection protrusion protruding from the first-first shield member, wherein the first-first connection protrusion protrudes from the insulation unit to be connected to a ground housing of a mating connector.
8. The board connector of claim 2, wherein the second ground contact includes a second-first ground contact located between a second-first RF contact among the second RF contacts and the transmit contacts with respect to the first axial direction, and a second-second ground contact located between a second-second RF contact among the second RF contacts and the transmit contacts with respect to the first axial direction, wherein the second-first ground contact includes a second-first shield member located between the second-first RF contact and the second-second RF contact with respect to the second axial direction.
9. The board connector of claim 8, wherein the first-first ground contact and the second-first ground contact are disposed to be point-symmetric with respect to a symmetry point that is spaced apart from each of both sidewalls of the ground housing, which are disposed to be spaced apart from each other with respect to the first axial direction, by the same distance, and spaced apart from each of both sidewalls of the ground housing, which are disposed to be spaced apart from each other with respect to the second axial direction, by the same distance, and the first-second ground contact and the second-second ground contact are disposed to be point-symmetric with respect to the symmetry point.
10. The board connector of claim 2, wherein the second ground contact includes a second-first ground contact formed in the same shape as the first-first ground contact, and a second-second ground contact formed in the same shape as the first-second ground contact.
11. The board connector of claim 1, wherein the ground housing includes a ground inner wall facing the insulation unit, a ground outer wall spaced apart from the ground inner wall, and a ground connection wall coupled to each of the ground inner wall and the ground outer wall, the ground inner wall includes a first sub-ground inner wall and a second sub-ground inner wall disposed to face each other with respect to the first axial direction, and a third sub-ground inner wall and a fourth sub-ground inner wall disposed to face each other with respect to the second axial direction, the first RF contacts are located between the first sub-ground inner wall and the first ground contact with respect to the first axial direction, and located between the third sub-ground inner wall and the fourth sub-ground inner wall with respect to the second axial direction, and the second RF contacts are located between the second sub-ground inner wall and the second ground contact with respect to the first axial direction, and located between the third sub-ground inner wall and the fourth sub-ground inner wall with respect to the second axial direction.
12. The board connector of claim 1, wherein the ground housing includes a ground inner wall facing the insulation unit, a ground outer wall spaced apart from the ground inner wall, and a ground connection wall coupled to each of the ground inner wall and the ground outer wall, the ground housing includes a conductive member coupled to an outer surface of the ground outer wall, and the conductive member is formed in a closed loop shape to extend along the ground outer wall, including a corner portion included in the ground outer wall.
13. The board connector of claim 1, wherein each of the first RF contacts includes a first RF mounting member in order to be mounted on a board, wherein each of the first RF mounting members is coupled to the insulation unit such that each of the first RF mounting members is located in a soldering inspection window that is formed by passing through the insulation unit.
14. The board connector of claim 1, wherein the ground housing includes a ground inner wall facing the insulation unit, a ground outer wall spaced apart from the ground inner wall and mounted on a board, and a ground connection wall coupled to each of the ground inner wall and the ground outer wall, and the ground housing is grounded through the ground outer wall mounted on

the board.

15. A board connector comprising: a plurality of radio frequency (RF) contacts for transmitting an RF signal; an insulation unit configured to support the RF contacts; a plurality of transmit contacts that are coupled to the insulation unit between a plurality of first RF contacts among the RF contacts and a plurality of second RF contacts among the RF contacts such that the first RF contacts and the second RF contacts are spaced apart from each other along a first axial direction; a ground housing to which the insulation unit is coupled; a first ground contact coupled to the insulation unit and configured to shield between the first RF contacts and the transmit contacts with respect to the first axial direction; and a second ground contact coupled to the insulation unit and configured to shield between the second RF contacts and the transmit contacts with respect to the first axial direction, wherein the ground housing comprises a ground sidewall surrounding a side of an inner side space and a ground bottom protruding toward the inner side space from a lower end of the ground sidewall, wherein the ground sidewall and the ground bottom have a seamless continuous surface without any gaps and are integrally formed, and wherein the ground housing is disposed to surround the first RF contacts and the second RF contacts arranged in the inner side space along the first axial direction and a second axial direction perpendicular to the first axial direction, such that the first RF contacts and the second RF contacts positioned in the inner space are completely shielded by the ground housing along the first axial direction and the second axial direction and are not exposed to an outside in the first axial direction and the second axial direction.

16. The board connector of claim 15, wherein a first-first RF contact among the first RF contacts and a first-second RF contact among the first RF contacts are spaced apart from each other along a second axial direction, the first ground contact is connected to a ground contact of a mating connector and realizes a shielding force that shields between the first-first RF contact and the first-second RF contact with respect to the second axial direction, and the ground housing includes: a first shield bottom protruding toward the first ground contact from the ground bottom so as to be located between the first-first RF contact and the first-second RF contact with respect to the second axial direction.

17. The board connector of claim 15, wherein a second-first RF contact among the second RF contacts and a second-second RF contact among the second RF contacts are spaced apart from each other along a second axial direction, the second ground contact is connected to a ground contact of a mating connector and realizes a shielding force that shields between the second-first RF contact and the second-second RF contact with respect to the second axial direction, and the ground housing includes: a second shield bottom protruding toward the second ground contact from the ground bottom so as to be located between the second-first RF contact and the second-second RF contact with respect to the second axial direction.

18. The board connector of claim 15, wherein the ground housing includes a conductive member coupled to an inner surface of the ground sidewall, wherein the conductive member is formed in a closed loop shape to extend along the inner surface of the ground sidewall, including a corner portion included in the inner surface of the ground sidewall.

19. The board connector of claim 15, wherein each of the first RF contacts includes a first RF mounting member in order to be mounted on a board, wherein each of the first RF mounting members is coupled to the insulation unit such that each of the first RF mounting members is located in a soldering inspection window that is formed by passing through the insulation unit.

20. The board connector of claim 15, wherein the ground housing is grounded through the ground bottom mounted on the board.